

Performance Analysis

[Reattempt](#)[Download Certificate](#)

Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

Answer Distribution

0

Correct

0

Incorrect

65

Unattempted

Domain Performance

Serverless	0%
Monitoring and Observability	0%
Security	0%
Database	0%
Compute	0%
Networking	0%
Storage	0%



Final Score

0/65

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes

Question Analysis

✓ Correct	0
✗ Incorrect	0
⊖ Unattempted	65
🔖 Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Not Attempted

Serverless



A company's marketing data is uploaded from multiple sources to an Amazon S3 bucket. A series of data preparation jobs aggregate the data for reporting. The data preparation jobs need to run at regular intervals in parallel. A subset of these jobs needs to run in a specific order later. The company wants to minimize the operational overhead of job error handling, retry logic, and state management. Which solution will meet these requirements?

- ☐ Use an AWS Lambda function to process the data as soon as the data is uploaded to the S3 bucket. Invoke other Lambda functions at regularly scheduled intervals.
Lambda functions alone are not well-suited for complex workflows requiring retry logic or specific execution order without significant custom coding.
- ☐ Use Amazon Athena to process the data. Use Amazon EventBridge Scheduler to invoke Athena on a regular interval.
Athena is a query service and does not inherently provide features for managing data preparation workflows, error handling, or job dependencies.
- ☒ Use AWS Glue DataBrew to process the data. Use an AWS Step Functions state machine to run the DataBrew data preparation jobs.
Glue DataBrew simplifies data preparation and integrates with Step Functions to manage workflows, error handling, and state management, including parallel and sequential job execution.
- ☐ Use AWS Data Pipeline to process the data. Schedule Data Pipeline to process the data once at midnight.
Data Pipeline requires significant manual configuration for error handling, retries, and dependency management, increasing operational overhead.

Overall Explanation:

AWS Glue DataBrew provides data preparation capabilities. Combining DataBrew with AWS Step Functions allows for defining and managing complex workflows, including parallel execution, sequential execution, error handling, retry logic, and state management, thereby minimizing operational overhead. Therefore, option C is the best solution.

[View Discussions](#)[Ask AI](#)

Question 2

Not Attempted

Serverless



A solutions architect is designing a payment processing application that runs on AWS Lambda in private subnets across multiple Availability Zones. The application uses multiple Lambda functions and processes millions of transactions each day. The architecture must ensure that the application does not process duplicate payments. Which solution will meet these requirements?

- ⊗ Use Lambda to retrieve all due payments. Publish the due payments to an Amazon S3 bucket. Configure the S3 bucket with an event notification to invoke another Lambda function to process the due payments.
S3 does not guarantee exactly-once processing, and using S3 events to trigger Lambda can result in duplicate invocations.
- ⊗ Use Lambda to retrieve all due payments. Publish the due payments to an Amazon Simple Queue Service (Amazon SQS) queue. Configure another Lambda function to poll the SQS queue and to process the due payments.
Standard SQS queues provide at-least-once delivery, which means duplicate messages are possible.
- ✔ Use Lambda to retrieve all due payments. Publish the due payments to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Configure another Lambda function to poll the FIFO queue and to process the due payments.
SQS FIFO queues provide exactly-once processing and maintain message order. This prevents duplicate payments and is suitable for high-throughput scenarios.
- ⊗ Use Lambda to retrieve all due payments. Store the due payments in an Amazon DynamoDB table. Configure streams on the DynamoDB table to invoke another Lambda function to process the due payments.
While DynamoDB Streams provide ordered delivery of changes, processing DynamoDB streams with Lambda can cause duplication.

Overall Explanation:

SQS FIFO queues are designed for exactly-once processing, which is essential to prevent duplicate payments. FIFO queues ensure each message is delivered only once, making them suitable for financial transactions. Also, standard queues provide at-least-once delivery, which means that each message is delivered at least once, meaning duplicates are possible. Using S3 or DynamoDB streams does not guarantee exactly-once processing.

[View Discussions](#)[Ask AI](#)

Question 3

Not Attempted

[Monitoring and Observability](#)

A company runs multiple workloads in its on-premises data center. The company's data center cannot scale fast enough to meet the company's expanding business needs. The company

wants to collect usage and configuration data about the on-premises servers and workloads to plan a migration to AWS. Which solution will meet these requirements?

- ☐ Set the home AWS Region in AWS Migration Hub. Use AWS Systems Manager to collect data about the on-premises servers.

Incorrect. While AWS Systems Manager can collect data, AWS Application Discovery Service is specifically designed for migration planning and integrates with AWS Migration Hub.

- ☒ Set the home AWS Region in AWS Migration Hub. Use AWS Application Discovery Service to collect data about the on-premises servers.

Correct. AWS Migration Hub provides a central location to track migrations. AWS Application Discovery Service is designed to collect the necessary usage and configuration data for migration planning.

- ☐ Use the AWS Schema Conversion Tool (AWS SCT) to create the relevant templates. Use AWS Trusted Advisor to collect data about the on-premises servers.

Incorrect. AWS SCT is primarily for database schema conversion, and AWS Trusted Advisor focuses on AWS resource optimization, not on-premises discovery.

- ☐ Use the AWS Schema Conversion Tool (AWS SCT) to create the relevant templates. Use AWS Database Migration Service (AWS DMS) to collect data about the on-premises servers.

Incorrect. AWS SCT is primarily for database schema conversion, and AWS DMS is for database migration, not on-premises discovery and assessment for general workload migration.

Overall Explanation:

AWS Application Discovery Service helps you plan your migration to the AWS cloud by collecting usage and configuration data about your on-premises servers and databases. AWS Migration Hub provides a central location to track the progress of your migration. Combining these two services provides the best solution for collecting the necessary data for planning a migration to AWS.

 View Discussions

 Ask AI

Question 4

Not Attempted

Security



A company has an organization in AWS Organizations with all features enabled. The company requires that all API calls and logins in any existing or new AWS account must be audited. The company needs a managed solution to prevent additional work and to minimize costs. The company also needs to know when any AWS account is not compliant with the AWS Foundational Security Best Practices (FSBP) standard. Which solution will meet these requirements with the LEAST operational overhead?

- ☒ Deploy an AWS Control Tower environment in the Organizations management account. Enable AWS Security Hub and AWS Control Tower Account Factory in the environment.

This is the correct answer. Control Tower provides centralized logging, monitoring, and account management. Security Hub monitors compliance with AWS Foundational Security Best Practices (FSBP), and Account Factory simplifies the creation of new accounts with preconfigured settings.

- ☐ Deploy an AWS Control Tower environment in a dedicated Organizations member account. Enable AWS Security Hub and AWS Control Tower Account Factory in the environment.

Control Tower is intended to be deployed in the management account, not a member account. This deviates from AWS-recommended practices.

- ☐ Use AWS Managed Services (AMS) Accelerate to build a multi-account landing zone (MALZ). Submit an RFC to self-service provision Amazon GuardDuty in the MALZ.

GuardDuty focuses on security threat detection rather than compliance monitoring for FSBP.

- ☐ Use AWS Managed Services (AMS) Accelerate to build a multi-account landing zone (MALZ). Submit an RFC to self-service provision AWS Security Hub in the MALZ.

AMS Accelerate is more suitable for enterprises with highly complex requirements and will likely incur more operational overhead and cost than the Control Tower solution. The question requires the solution with the least operational overhead and the minimum cost.

Overall Explanation:

AWS Control Tower in the management account with Security Hub and Account Factory provides a managed, centralized, and cost-effective solution for auditing and compliance monitoring across all AWS accounts within an organization. It automates account provisioning, enforces security policies, and provides a dashboard for monitoring compliance with AWS Foundational Security Best Practices (FSBP). Security Hub integrated with Control Tower provides a single pane of glass for managing your security posture across multiple accounts and regions .

[View Discussions](#)[Ask AI](#)

Question 5

Not Attempted

Database



A company has stored 10 TB of log files in Apache Parquet format in an Amazon S3 bucket. The company occasionally needs to use SQL to analyze the log files. Which solution will meet these requirements MOST cost-effectively?

- ☐ Create an Amazon Aurora MySQL database. Migrate the data from the S3 bucket into Aurora by using AWS Database Migration Service (AWS DMS). Issue SQL statements to the Aurora database.

Migrating 10 TB of data incurs significant costs and operational overhead.

- ☐ Create an Amazon Redshift cluster. Use Redshift Spectrum to run SQL statements directly on the data in the S3 bucket.

Redshift Spectrum allows querying data directly in S3 without loading it into Redshift, but costs are high, especially for infrequent use.

- ☒ Create an AWS Glue crawler to store and retrieve table metadata from the S3 bucket. Use Amazon Athena to run SQL statements directly on the data in the S3 bucket.

Athena is serverless and charges only for the data scanned by queries. Glue Crawler automatically extracts metadata and schema information from the Parquet files. No data migration is needed.

- ☐ Create an Amazon EMR cluster. Use Apache Spark SQL to run SQL statements directly on the data in the S3 bucket.

Using EMR for infrequent queries would incur high costs.

Overall Explanation:

Amazon Athena, combined with AWS Glue for schema discovery, provides a cost-effective solution for querying data in S3 using SQL. Athena is serverless, meaning you only pay for the queries you run, making it ideal for occasional analysis. Glue efficiently catalogs the Parquet data without requiring data migration. Therefore, it is the most cost-effective option. Redshift and EMR are more expensive solutions. Aurora would also incur significant costs to migrate the data.

[View Discussions](#)[Ask AI](#)

Question 6

Not Attempted

Security



A company needs a solution to prevent AWS CloudFormation stacks from deploying AWS Identity and Access Management (IAM) resources that include an inline policy or "*" in the statement. The solution must also prohibit deployment of Amazon EC2 instances with public IP addresses. The company has AWS Control Tower enabled in its organization in AWS Organizations. Which solution will meet these requirements?

- ☒ Use AWS Control Tower proactive controls to block deployment of EC2 instances with public IP addresses and inline policies with elevated access or "*".
Proactive controls, implemented using AWS CloudFormation hooks, prevent the creation of non-compliant resources by validating configurations before deployment.
- ☐ Use AWS Control Tower detective controls to block deployment of EC2 instances with public IP addresses and inline policies with elevated access or "*".
Detective controls detect specific events after they occur, they cannot prevent the initial deployment.
- ☐ Use AWS Config to create rules for EC2 and IAM compliance. Configure the rules to run an AWS Systems Manager Session Manager automation to delete a resource when it is not compliant.
AWS Config rules detect non-compliance but do not prevent deployment. Automatically deleting resources is a reactive, not proactive, measure.
- ☐ Use a service control policy (SCP) to block actions for the EC2 instances and IAM resources if the actions lead to noncompliance.
SCPs can indeed block specific API calls for creating IAM resources with "*" or EC2 instances with public IPs, but proactive control of AWS Control Tower that makes the most of AWS CloudFormation stacks is a better approach.

Overall Explanation:

AWS Control Tower proactive controls are the most appropriate solution. They prevent the deployment of non-compliant resources by using AWS CloudFormation hooks to validate configurations before deployment. This directly addresses the requirements of blocking IAM resources with inline policies containing "*" and EC2 instances with public IP addresses. Detective controls only identify non-compliance after deployment. AWS Config detects violations, but it's reactive. SCPs are a valid preventive approach, but Control Tower proactive controls offer better integration with CloudFormation and more granular control

at the CloudFormation stack level for these specific compliance rules. Links for both * in inline policy: <https://docs.aws.amazon.com/controltower/latest/userguide/iam-rules.html#ct-iam-pr-1-description> and for ec2 public IP: <https://docs.aws.amazon.com/controltower/latest/userguide/ec2-rules.html#ct-ec2-pr-9-description>

[View Discussions](#)[Ask AI](#)

Question 7

Not Attempted

Database



A company is planning to use an Amazon DynamoDB table for data storage and is concerned about cost optimization. The table will not be used on most mornings. In the evenings, the read and write traffic will often be unpredictable and experience quick spikes. What DynamoDB capacity mode and configuration should a solutions architect recommend to achieve cost optimization while handling these traffic patterns?



Create a DynamoDB table in on-demand capacity mode.

On-demand mode is suitable for unpredictable traffic patterns and infrequent use, as it charges only for actual usage. This aligns with the table being mostly unused in the mornings and experiencing unpredictable spikes in the evenings. It avoids the need to provision and potentially overpay for unused capacity.



Create a DynamoDB table with a global secondary index.

A global secondary index is used to optimize query performance for different access patterns, but it doesn't address the core requirement of cost optimization in the face of unpredictable traffic. While useful for performance, it adds complexity and cost if not directly related to optimizing for the described usage patterns.



Create a DynamoDB table with provisioned capacity and auto scaling.

While provisioned capacity with auto scaling can handle traffic spikes, it may not be the most cost-effective solution for workloads with extended periods of inactivity, such as the table being unused most mornings. Auto scaling takes time to adjust, potentially leading to throttling during initial spikes, and requires setting minimum capacity levels that might incur unnecessary costs during low-usage periods.



Create a DynamoDB table in provisioned capacity mode, and configure it as a global table.

Configuring a table as a global table replicates it across multiple AWS regions, which is primarily for disaster recovery and low-latency access in different geographic locations. This adds significant cost

and complexity and doesn't directly address the requirement of cost optimization for unpredictable traffic patterns within a single region.

Overall Explanation:

The best option for cost optimization with unpredictable traffic and periods of inactivity is on-demand capacity mode. This mode automatically scales to handle traffic spikes without any pre-configured capacity, and you only pay for the read and write operations that your application performs. Provisioned capacity with auto-scaling can be a good fit for predictable workloads. Global secondary indexes are for query optimization and global tables are for disaster recovery. Neither of these are relevant in this scenario.

[View Discussions](#)[Ask AI](#)

Question 8

Not Attempted

[Compute](#)

A company's web application, hosted in the AWS Cloud, has recently experienced a surge in popularity. Currently, the web application resides on a single Amazon EC2 instance within a single public subnet and struggles to handle the increased web traffic demand. The company requires a solution providing high availability and scalability to address the escalating user demand without necessitating a rewrite of the existing web application. Which combination of actions will effectively meet these requirements? (Choose two.)

- ☐ Replace the EC2 instance with a larger compute optimized instance.
While a larger instance might offer temporary relief, it doesn't address high availability or automatic scaling.
- ☒ Configure Amazon EC2 Auto Scaling with multiple Availability Zones in private subnets.
EC2 Auto Scaling dynamically adjusts the number of EC2 instances based on demand, ensuring scalability and high availability across multiple Availability Zones.
- ☐ Configure a NAT gateway in a public subnet to handle web requests.
A NAT gateway enables instances in private subnets to connect to the internet but doesn't address scalability or load distribution for incoming web traffic.

- ☐ Replace the EC2 instance with a larger memory optimized instance.
Similar to compute-optimized instances, memory-optimized instances don't inherently provide high availability or automatic scaling.
- ☒ Configure an Application Load Balancer in a public subnet to distribute web traffic.
An Application Load Balancer distributes incoming web traffic across multiple EC2 instances, enhancing performance and availability. It handles traffic distribution and health checks.

Overall Explanation:

The optimal solution involves utilizing both EC2 Auto Scaling and an Application Load Balancer. EC2 Auto Scaling (option B) ensures that the application can scale up or down based on demand across multiple Availability Zones, enhancing availability and fault tolerance. An Application Load Balancer (option E) distributes incoming traffic across these instances, providing a single point of contact for users and improving application performance. This combination achieves high availability and scalability without requiring code changes.

[View Discussions](#)[Ask AI](#)

Question 9

Not Attempted

Security



A company develops AWS Lambda functions that rely on environment variables. For security reasons, the company mandates that developers should not be able to view these environment variables in plaintext. Which of the following solutions fulfills this requirement?

- ☐ Deploy the application code to Amazon EC2 instances instead of using Lambda functions.
This option moves away from Lambda functions entirely, which might not be desirable and doesn't directly address the environment variable security concern.
- ☐ Configure SSL encryption on the Lambda functions and use AWS CloudHSM to store and encrypt the environment variables.
SSL encryption is for data in transit, not data at rest (environment variables). CloudHSM is an option but is not the ideal service for encrypting Lambda environment variables.

- ⊗ Create a certificate in AWS Certificate Manager (ACM). Configure the Lambda functions to use the certificate to encrypt the environment variables.

ACM certificates are used for securing network traffic (TLS/SSL), not for encrypting environment variables at rest.

- ✔ Create an AWS Key Management Service (AWS KMS) key. Enable encryption helpers on the Lambda functions to use the KMS key to store and encrypt the environment variables.

This is the correct solution. AWS KMS is designed for managing encryption keys, and Lambda's encryption helpers make it easy to encrypt environment variables using a KMS key, protecting them from being viewed in plaintext.

Overall Explanation:

AWS KMS allows you to encrypt your environment variables at rest. Lambda encryption helpers are the best choice to encrypt the variables using KMS, preventing developers from viewing them in plaintext.

 View Discussions

 Ask AI

Question 10

Not Attempted

Security



An analytics company runs its multi-tier services within an Amazon VPC. The company aims to provide a web analytics service, accessible via RESTful APIs, to millions of users. To ensure security, all users must be authenticated before accessing the APIs. Which solution achieves these requirements with the HIGHEST operational efficiency?

- ✔ Configure an Amazon Cognito user pool for user authentication. Implement Amazon API Gateway REST APIs with a Cognito authorizer.

Using a Cognito user pool handles authentication and user management efficiently. API Gateway's Cognito authorizer natively integrates with the user pool to secure the REST APIs.

- ⊗ Configure an Amazon Cognito identity pool for user authentication. Implement Amazon API Gateway HTTP APIs with a Cognito authorizer.

Identity pools are primarily for authorization, granting access to AWS resources, not user authentication. HTTP APIs offer lower cost, but might not be the best choice for this scenario. User pools are generally preferred for authentication over Identity Pools.

- ⊗ Configure an AWS Lambda function to handle user authentication. Implement Amazon API Gateway REST APIs with a Lambda authorizer.

While a Lambda authorizer is possible, it adds operational overhead compared to using Cognito user pools directly, as it requires you to manage the entire authentication process within the Lambda function. This increases complexity and reduces operational efficiency.

- ⊗ Configure an IAM user to handle user authentication. Implement Amazon API Gateway HTTP APIs with an IAM authorizer.

Using IAM users for direct user authentication is not recommended for web applications due to security concerns and operational overhead. This approach is less scalable and harder to manage than using a dedicated user management service like Cognito. It's also an anti-pattern to expose IAM credentials directly to users.

Overall Explanation:

Amazon Cognito user pools provide a managed user directory for authentication.

Integrating this with API Gateway via a Cognito authorizer offers a scalable, secure, and operationally efficient solution for verifying users before they access RESTful APIs. This eliminates the need to manage custom authentication logic within Lambda functions or directly expose IAM credentials.

 View Discussions

 Ask AI

Question 11

Not Attempted

Security



A company has a mobile app for customers. The app's data is sensitive and must be encrypted at rest. The company uses AWS Key Management Service (AWS KMS). The company needs a solution that prevents the accidental deletion of KMS keys. The solution must use Amazon Simple Notification Service (Amazon SNS) to send an email notification to administrators when a user attempts to delete a KMS key. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Create an Amazon EventBridge rule that reacts when a user tries to delete a KMS key. Configure an AWS Config rule that cancels any deletion of a KMS key. Add the AWS Config rule as a target of the EventBridge rule. Create an SNS topic that notifies the administrators.

AWS Config is primarily for compliance monitoring, not real-time prevention of actions. It's less suited for immediate cancellation of KMS key deletions and has higher operational overhead than

other solutions.

- ⊗ Create an AWS Lambda function that has custom logic to prevent KMS key deletion. Create an Amazon CloudWatch alarm that is activated when a user tries to delete a KMS key. Create an Amazon EventBridge rule that invokes the Lambda function when the DeleteKey operation is performed. Create an SNS topic. Configure the EventBridge rule to publish an SNS message that notifies the administrators.

Using a Lambda function adds unnecessary complexity and operational overhead. CloudWatch alarms are also not the most efficient way to prevent the deletion. There is a simpler, more direct method.

- ✓ Create an Amazon EventBridge rule that reacts when the KMS DeleteKey operation is performed. Configure the rule to initiate an AWS Systems Manager Automation runbook. Configure the runbook to cancel the deletion of the KMS key. Create an SNS topic. Configure the EventBridge rule to publish an SNS message that notifies the administrators.

This is the most efficient solution. EventBridge directly monitors the DeleteKey operation, triggering a Systems Manager Automation runbook to cancel the deletion. This provides a direct and automated way to prevent accidental deletion while also notifying administrators via SNS.

- ⊗ Create an AWS CloudTrail trail. Configure the trail to deliver logs to a new Amazon CloudWatch log group. Create a CloudWatch alarm based on the metric filter for the CloudWatch log group. Configure the alarm to use Amazon SNS to notify the administrators when the KMS DeleteKey operation is performed.

CloudTrail is for auditing, not immediate prevention. While CloudWatch alarms can notify administrators, this solution does not prevent the key deletion from proceeding, and it requires more setup than option C.

Overall Explanation:

The most efficient solution is to use Amazon EventBridge to react to the KMS DeleteKey operation and trigger an AWS Systems Manager Automation runbook to cancel the deletion. This approach minimizes operational overhead by directly addressing the requirement with readily available services, and also sending the notifications via SNS.

 View Discussions

 Ask AI

Question 12

Not Attempted

Compute



A company wants to analyze and generate reports to track the usage of its mobile app. The app is popular and has a global user base. The company uses a custom report building program to analyze application usage. The program generates multiple reports during the last week of each month. The program takes less than 10 minutes to produce each report. The company rarely uses the program to generate reports outside of the last week of each month. The company wants to generate reports in the least amount of time when the reports are requested. Which solution will meet these requirements MOST cost-effectively?

- ☐ Run the program by using Amazon EC2 On-Demand Instances. Create an Amazon EventBridge rule to start the EC2 instances when reports are requested. Run the EC2 instances continuously during the last week of each month.

Running EC2 instances continuously, even On-Demand, is less cost-effective than other serverless options when the program only runs briefly on request. It involves managing infrastructure.

- ☒ Run the program in AWS Lambda. Create an Amazon EventBridge rule to run a Lambda function when reports are requested.

AWS Lambda is a cost-effective solution for running short-duration tasks on demand. EventBridge can trigger the Lambda function when reports are requested. Since the reports only take 10 minutes to generate, and Lambda supports up to 15 minutes execution time, this is a good fit. Lambda also requires minimal operational overhead.

- ☐ Run the program in Amazon Elastic Container Service (Amazon ECS). Schedule Amazon ECS to run the program when reports are requested.

ECS can run the reports, but it is less cost-effective and more complex to manage than Lambda for this particular use case, where the execution time is short and infrequent.

- ☐ Run the program by using Amazon EC2 Spot Instances. Create an Amazon EventBridge rule to start the EC2 instances when reports are requested. Run the EC2 instances continuously during the last week of each month.

While Spot Instances are cheaper than On-Demand, running them continuously during the last week of the month is not cost-effective. Additionally, Spot Instances can be interrupted, causing potential delays in report generation. It also involves more operational overhead than serverless options.

Overall Explanation:

AWS Lambda is the most cost-effective solution because it runs the program only when needed, it avoids the cost of running EC2 instances continuously, and it requires minimal operational overhead. The custom report building program takes less than 10 minutes to generate reports and the maximum Lambda execution time is 15 minutes. EventBridge

triggers Lambda based on report requests. Therefore Lambda is the best serverless compute choice.

[View Discussions](#)[Ask AI](#)

Question 13

Not Attempted

Networking



A company is designing a tightly coupled high-performance computing (HPC) environment on AWS. The company needs to optimize the HPC environment for networking and storage to ensure low latency and high throughput. Which combination of solutions will meet these requirements? (Choose two.)

- ☐ Create an accelerator in AWS Global Accelerator. Configure custom routing for the accelerator.

AWS Global Accelerator is designed for improving performance for geographically distributed users, not for low-latency intra-cluster communication in tightly coupled HPC environments.

- ☒ Create an Amazon FSx for Lustre file system. Configure the file system with scratch storage.

Amazon FSx for Lustre is specifically designed for HPC workloads, providing high-throughput and low-latency access to data. Using it as scratch storage is ideal for HPC.

- ☐ Create an Amazon CloudFront distribution. Configure the viewer protocol policy to be HTTP and HTTPS.

Amazon CloudFront is a content delivery network (CDN) and does not address the low-latency, high-bandwidth networking or storage requirements of tightly coupled HPC environments.

- ☒ Launch Amazon EC2 instances. Attach an Elastic Fabric Adapter (EFA) to the instances.

Elastic Fabric Adapter (EFA) provides low-latency, high-throughput connectivity for tightly coupled HPC workloads, especially those using MPI (Message Passing Interface).

- ☐ Create an AWS Elastic Beanstalk deployment to manage the environment.

AWS Elastic Beanstalk is used for deploying and managing web applications and services and is not suitable for HPC environments.

Overall Explanation:

Amazon FSx for Lustre provides the high-performance storage necessary for HPC workloads. Elastic Fabric Adapter (EFA) provides the necessary low-latency networking for tightly coupled HPC clusters, enabling high-speed inter-node communication using MPI.

[View Discussions](#)[Ask AI](#)

Question 14

Not Attempted

Security



A company needs a solution to prevent photos with unwanted content from being uploaded to the company's web application. The solution must NOT involve training a machine learning (ML) model.

- ☐ Create and deploy a model by using Amazon SageMaker Autopilot. Create a real-time endpoint that the web application invokes when new photos are uploaded.

This option involves training an ML model using SageMaker Autopilot, which violates the requirement.

- ☒ Create an AWS Lambda function that uses Amazon Rekognition to detect unwanted content. Create a Lambda function URL that the web application invokes when new photos are uploaded.

This option uses Amazon Rekognition, which is a pre-trained service for image and video analysis and doesn't require training a custom model. It meets the requirements.

- ☐ Create an Amazon CloudFront function that uses Amazon Comprehend to detect unwanted content. Associate the function with the web application.

Amazon Comprehend is a natural language processing (NLP) service and is not designed for analyzing images. Therefore, this option is not suitable.

- ☐ Create an AWS Lambda function that uses Amazon Rekognition Video to detect unwanted content. Create a Lambda function URL that the web application invokes when new photos are uploaded.

While Rekognition Video can analyze videos for unwanted content, the requirement is for photos, making Rekognition Image a better fit if choosing Rekognition. However, the core logic of using Rekognition to avoid custom model training is present, although not perfectly tailored. Option B is still more relevant.

Overall Explanation:

The correct answer is to use an AWS Lambda function with Amazon Rekognition. Rekognition is a pre-trained service that can analyze images and detect unwanted content without requiring the training of a custom ML model, fulfilling the requirements. Comprehend is for NLP, and SageMaker Autopilot requires training a model.

[View Discussions](#)[Ask AI](#)

Question 15

Not Attempted

Security



A company's critical ecommerce platform runs on AWS with high traffic and transaction volumes. They've secured the AWS account root user with an MFA device. To prevent losing access to the root user account if the MFA device is lost, what is the BEST solution?

- ☐ Set up a backup administrator IAM user that the company can use if the MFA device is lost.

Creating a backup IAM user provides an alternative administrative access point. However, it doesn't directly address the root user account access issue.

- ☒ Add multiple MFA devices for the root user account to handle the disaster scenario.

Adding multiple MFA devices to the root user account provides redundancy. If one device is lost or unavailable, another can be used for authentication, ensuring continuous access.

- ☐ Create a new administrator account when the company cannot access the root account.

Creating a new administrator account after losing access is not a proactive solution and involves a recovery process instead of preventing the access loss.

- ☐ Attach the administrator policy to another IAM user when the company cannot access the root account.

Attaching the administrator policy to another IAM user is a reactive measure and does not prevent the initial loss of access. It also violates the principle of least privilege.

Overall Explanation:

Adding multiple MFA devices for the root user account is the best solution. It provides a built-in redundancy mechanism. If one MFA device is lost, the company can still access the root user account using another registered MFA device. AWS best practices recommend this approach for root user accounts to ensure business continuity and security.

[View Discussions](#)[Ask AI](#)

Question 16

Not Attempted

Serverless



A social media company is creating a rewards program website for its users. The company gives users points when users create and upload videos to the website. Users redeem their points for gifts or discounts from the company's affiliated partners. A unique ID identifies users. The partners refer to this ID to verify user eligibility for rewards. The partners want to receive notification of user IDs through an HTTP endpoint when the company gives users points. Hundreds of vendors are interested in becoming affiliated partners every day. The company wants to design an architecture that gives the website the ability to add partners rapidly in a scalable way. Which solution will meet these requirements with the LEAST implementation effort?

- ☐ Create an Amazon Timestream database to keep a list of affiliated partners. Implement an AWS Lambda function to read the list. Configure the Lambda function to send user IDs to each partner when the company gives users points.

This approach requires more implementation effort than necessary. It involves managing a database and configuring a Lambda function to handle notifications, which is less efficient than using a managed service designed for this purpose.

- ☒ Create an Amazon Simple Notification Service (Amazon SNS) topic. Choose an endpoint protocol. Subscribe the partners to the topic. Publish user IDs to the topic when the company gives users points.

SNS is designed for this type of use case. It allows publishing messages to a topic and delivering them to multiple subscribers. Partners can subscribe using an HTTP endpoint. This is scalable and requires minimal implementation effort.

- ☐ Create an AWS Step Functions state machine. Create a task for every affiliated partner. Invoke the state machine with user IDs as input when the company gives users points.

Step Functions would involve more overhead than necessary. Managing a state machine with individual tasks for each partner adds complexity and is not the most efficient solution for simple notifications.

- ⊗ Create a data stream in Amazon Kinesis Data Streams. Implement producer and consumer applications. Store a list of affiliated partners in the data stream. Send user IDs when the company gives users points.

Kinesis Data Streams is designed for real-time data streaming and is an overkill for this scenario. Storing the list of partners in the data stream adds unnecessary complexity.

Overall Explanation:

Amazon SNS is the best solution because it is designed for sending notifications to multiple subscribers using a publish/subscribe model. Partners can subscribe to the SNS topic via HTTP endpoints, providing a scalable and low-effort solution for notifying partners when users receive points. Options A, C, and D are more complex and require more implementation effort.

 View Discussions

 Ask AI

Question 17

Not Attempted

Serverless



A company needs to extract the names of ingredients from recipe records that are stored as text files in an Amazon S3 bucket. A web application will use the ingredient names to query an Amazon DynamoDB table and determine a nutrition score. The application can handle non-food records and errors. The company does not have any employees who have machine learning knowledge to develop this solution. Which solution will meet these requirements MOST cost-effectively?

- ✔ Use S3 Event Notifications to invoke an AWS Lambda function when PutObject requests occur. Program the Lambda function to analyze the object and extract the ingredient names by using Amazon Comprehend. Store the Amazon Comprehend output in the DynamoDB table.

Amazon Comprehend is a natural language processing (NLP) service that can be used to extract entities like ingredient names from text. Lambda provides a serverless compute environment to process the S3 events.

- ⊗ Use an Amazon EventBridge rule to invoke an AWS Lambda function when PutObject requests occur. Program the Lambda function to analyze the object by using Amazon Forecast to extract the ingredient names. Store the Forecast output in the DynamoDB table.

Amazon Forecast is designed for time-series forecasting, not for extracting entities from text. This makes it unsuitable for the task.

- ⊗ Use S3 Event Notifications to invoke an AWS Lambda function when PutObject requests occur. Use Amazon Polly to create audio recordings of the recipe records. Save the audio files in the S3 bucket. Use Amazon Simple Notification Service (Amazon SNS) to send a URL as a message to employees. Instruct the employees to listen to the audio files and calculate the nutrition score. Store the ingredient names in the DynamoDB table.

This solution is inefficient and requires manual intervention from employees, making it less cost-effective and scalable.

- ⊗ Use an Amazon EventBridge rule to invoke an AWS Lambda function when a PutObject request occurs. Program the Lambda function to analyze the object and extract the ingredient names by using Amazon SageMaker. Store the inference output from the SageMaker endpoint in the DynamoDB table.

Amazon SageMaker is a machine learning service. The company does not have employees with ML knowledge, and it's overkill for this task. Also, Amazon Comprehend provides similar features with less setup.

Overall Explanation:

The most cost-effective solution is to use S3 Event Notifications to trigger a Lambda function that uses Amazon Comprehend to extract ingredient names. This approach leverages a managed NLP service that requires no ML expertise from the company. It's also serverless and scalable. Options B and D involve services not suited for the task or requiring ML expertise, and Option C requires manual intervention, making them less efficient and cost-effective.

[View Discussions](#)[Ask AI](#)

Question 18

Not Attempted

[Compute](#)

A company is migrating a distributed application to AWS. The application serves variable workloads. The legacy platform consists of a primary server that coordinates jobs across multiple compute nodes. The company wants to modernize the application with a solution that maximizes resiliency and scalability. How should a solutions architect design the architecture to meet these requirements?

- ⊗ Configure an Amazon Simple Queue Service (Amazon SQS) queue as a destination for the jobs. Implement the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure EC2 Auto Scaling to use scheduled scaling.

Scheduled scaling does not adapt well to variable workloads as it relies on predefined schedules rather than real-time demand.

- ✔ Configure an Amazon Simple Queue Service (Amazon SQS) queue as a destination for the jobs. Implement the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure EC2 Auto Scaling based on the size of the queue.

Using SQS decouples the primary server from the compute nodes, enhancing resiliency and scalability. Scaling the Auto Scaling group based on queue size ensures that compute capacity dynamically adjusts to the workload.

- ⊗ Implement the primary server and the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure AWS CloudTrail as a destination for the jobs. Configure EC2 Auto Scaling based on the load on the primary server.

CloudTrail is designed for auditing API calls, not for managing job destinations. Furthermore, placing the primary server and compute nodes in the same Auto Scaling group can limit scalability.

- ⊗ Implement the primary server and the compute nodes with Amazon EC2 instances that are managed in an Auto Scaling group. Configure Amazon EventBridge (Amazon CloudWatch Events) as a destination for the jobs. Configure EC2 Auto Scaling based on the load on the compute nodes.

EventBridge is not suitable for direct job management. Scaling the entire Auto Scaling group based solely on compute node load does not provide a decoupled and scalable solution.

Overall Explanation:

The optimal solution involves using Amazon SQS to decouple the job queue from the compute nodes. This allows the compute nodes, managed by an Auto Scaling group, to independently scale based on the size of the SQS queue. This approach maximizes resiliency, scalability, and cost-efficiency by dynamically adjusting resources to meet variable workload demands. Option B is the only solution offering this.

 View Discussions

 Ask AI

Question 19

Not Attempted

Security



A company recently partnered with an AWS Managed Service Provider (MSP) for an application migration. A solutions architect needs to share an Amazon Machine Image (AMI) from the company's AWS account with the MSP's AWS account. The AMI is backed by Amazon Elastic Block Store (Amazon EBS) and uses an AWS Key Management Service (AWS KMS) customer managed key to encrypt the EBS volume snapshots. What is the MOST secure way to share the AMI with the MSP's AWS account?

- ☐ **Make the encrypted AMI and snapshots publicly available. Modify the key policy to allow the MSP's AWS account to use the key.**
Making the AMI and snapshots public is not secure as it would allow anyone to use them.
- ☒ **Modify the launchPermission property of the AMI. Share the AMI with the MSP's AWS account only. Modify the key policy to allow the MSP's AWS account to use the key.**
Modifying the launchPermission property and sharing the AMI with the MSP's account only, along with granting the MSP access to the KMS key, is the most secure option.
- ☐ **Modify the launchPermission property of the AMI. Share the AMI with the MSP's AWS account only. Modify the key policy to trust a new KMS key that is owned by the MSP for encryption.**
Trusting a new KMS key owned by the MSP doesn't grant them access to the existing key used to encrypt the AMI. The original key is required to decrypt the snapshots.
- ☐ **Export the AMI from the source account to an Amazon S3 bucket in the MSP's AWS account. Encrypt the S3 bucket with a new KMS key that is owned by the MSP. Copy and launch the AMI in the MSP's AWS account.**
Exporting the AMI to an S3 bucket will still require the original KMS key to decrypt the AMI's contents. This option doesn't solve the KMS key access issue.

Overall Explanation:

The most secure way is to modify the AMI's launch permissions to only allow the MSP's AWS account to launch instances from it. It's also crucial to modify the KMS key policy to grant the MSP's AWS account permission to use the KMS key for decryption. This ensures only the MSP can use the AMI and they have the necessary permissions to access the encrypted EBS snapshots.

[View Discussions](#)[Ask AI](#)

Question 20

Not Attempted

Serverless



A company needs to create an AWS Lambda function that will run in a VPC in the company's primary AWS account. The Lambda function needs to access files that the company stores in an Amazon Elastic File System (Amazon EFS) file system. The EFS file system is located in a secondary AWS account. As the company adds files to the file system, the solution must scale to meet the demand. Which solution will meet these requirements MOST cost-effectively?

- ☐ Create a new EFS file system in the primary account. Use AWS DataSync to copy the contents of the original EFS file system to the new EFS file system.
Duplicating the EFS file system leads to unnecessary storage costs and adds operational complexity due to the need for data synchronization. It's not the most cost-effective solution.
- ☒ Create a VPC peering connection between the VPCs that are in the primary account and the secondary account.
A VPC peering connection allows the Lambda function in the primary account to directly access the EFS file system in the secondary account securely and efficiently. This approach avoids data duplication or complex function invocations and scales with demand.
- ☐ Create a second Lambda function in the secondary account that has a mount that is configured for the file system. Use the primary account's Lambda function to invoke the secondary account's Lambda function.
Creating a second Lambda function adds unnecessary complexity to the architecture. Invoking Lambda functions across accounts introduces additional latency and cost compared to a direct connection.
- ☐ Move the contents of the file system to a Lambda layer. Configure the Lambda layer's permissions to allow the company's secondary account to use the Lambda layer.
Lambda layers are designed for sharing code and dependencies, not for storing large amounts of data like an entire file system. This option is not practical or cost-effective.

Overall Explanation:

VPC peering allows the Lambda function in the primary AWS account to directly access the EFS file system in the secondary AWS account. This solution is the most cost-effective because it avoids data duplication (Option A), unnecessary function invocations (Option C), and misuse of Lambda layers for data storage (Option D). VPC peering enables a secure

and efficient connection between the two accounts for accessing the EFS file system and scales with the data being accessed.

[View Discussions](#)[Ask AI](#)

Question 21

Not Attempted

Security



A financial company needs to handle highly sensitive data and will store it in an Amazon S3 bucket. The company requires that the data be encrypted both in transit and at rest. Critically, the company must manage the encryption keys outside the AWS Cloud. Which solution will meet these requirements?

- ☐ Encrypt the data in the S3 bucket with server-side encryption (SSE) that uses an AWS Key Management Service (AWS KMS) customer managed key.
This option requires using AWS KMS, thus the keys are managed within AWS Cloud, which does not meet the requirement.
- ☐ Encrypt the data in the S3 bucket with server-side encryption (SSE) that uses an AWS Key Management Service (AWS KMS) AWS managed key.
This option uses AWS KMS AWS managed keys, meaning that the customer does not manage the keys outside of the AWS cloud, which is against the requirements.
- ☐ Encrypt the data in the S3 bucket with the default server-side encryption (SSE).
Default server-side encryption keys are managed by AWS. This does not meet the requirement of managing keys outside of AWS.
- ☒ Encrypt the data at the company's data center before storing the data in the S3 bucket.
Encrypting the data at the company's data center (client-side encryption) allows the company to manage the encryption keys outside of the AWS Cloud. It also encrypts the data in transit and at rest.

Overall Explanation:

Encrypting the data at the company's data center before sending it to S3 satisfies both the encryption in transit and at rest requirements, while also allowing the company to manage encryption keys outside the AWS Cloud. This approach is known as client-side encryption.

[View Discussions](#)[Ask AI](#)

Question 22

Not Attempted

Serverless



A company is building a payment application on AWS that receives payment notifications from mobile devices. These notifications require basic validation before further processing. The backend application for processing is long-running and needs compute and memory to be adjusted dynamically. The company wants to minimize operational overhead and avoid managing infrastructure. Which solution best meets these requirements?

- ☐ Create an Amazon Simple Queue Service (Amazon SQS) queue. Integrate the queue with an Amazon EventBridge rule to receive payment notifications from mobile devices. Configure the rule to validate payment notifications and send the notifications to the backend application. Deploy the backend application on Amazon Elastic Kubernetes Service (Amazon EKS) Anywhere. Create a standalone cluster.
EKS Anywhere involves managing on-premises infrastructure, which contradicts the requirement to avoid infrastructure management and minimize operational overhead. EventBridge rule is not meant for data validation.
- ☐ Create an Amazon API Gateway API. Integrate the API with an AWS Step Functions state machine to receive payment notifications from mobile devices. Invoke the state machine to validate payment notifications and send the notifications to the backend application. Deploy the backend application on Amazon Elastic Kubernetes Service (Amazon EKS). Configure an EKS cluster with self-managed nodes.
While EKS offers scalability, self-managed nodes require infrastructure management, increasing operational overhead. Step functions are also an overkill for basic validation.
- ☐ Create an Amazon Simple Queue Service (Amazon SQS) queue. Integrate the queue with an Amazon EventBridge rule to receive payment notifications from mobile devices. Configure the rule to validate payment notifications and send the notifications to the backend application. Deploy the backend application on Amazon EC2 Spot Instances. Configure a Spot Fleet with a default allocation strategy.
Using EC2 Spot Instances and Spot Fleets still involves managing EC2 instances and dealing with potential interruptions, which increases operational overhead. EventBridge rule is not meant for data validation.
- ☒ Create an Amazon API Gateway API. Integrate the API with AWS Lambda to receive payment notifications from mobile devices. Invoke a Lambda function to validate payment notifications and send the notifications to the backend application. Deploy the

backend application on Amazon Elastic Container Service (Amazon ECS). Configure Amazon ECS with an AWS Fargate launch type.

This solution provides a serverless architecture. API Gateway handles API requests, Lambda performs validation, and ECS with Fargate manages the backend application with no infrastructure management. Fargate automatically scales resources.

Overall Explanation:

The optimal solution is option D. API Gateway and Lambda offer serverless capabilities for receiving and validating payment notifications, minimizing operational overhead. ECS with Fargate provides a managed container service that automatically scales compute and memory resources without requiring infrastructure management. Options A, B, and C all involve managing infrastructure components, which increases operational overhead.

[View Discussions](#)[Ask AI](#)

Question 23

Not Attempted

Security



A solutions architect is designing a user authentication solution for a company. The solution must enforce two-factor authentication (2FA) for users that log in from inconsistent geographical locations, IP addresses, or devices. The solution must also be able to scale up to accommodate millions of users. Which solution will meet these requirements?

- ☒ **Configure Amazon Cognito user pools for user authentication. Enable the risk-based adaptive authentication feature with multifactor authentication (MFA).**
Amazon Cognito user pools, with risk-based adaptive authentication and MFA, can enforce 2FA based on factors like location, IP address, and device. It also scales to millions of users. Adaptive authentication assigns a risk score to user activity and can require MFA based on suspicious activity.
- ☐ **Configure Amazon Cognito identity pools for user authentication. Enable multi-factor authentication (MFA).**
Amazon Cognito identity pools are used for granting temporary AWS credentials to users. They do not inherently support adaptive authentication or risk-based MFA based on login characteristics.

- ⊗ Configure AWS Identity and Access Management (IAM) users for user authentication. Attach an IAM policy that allows the AllowManageOwnUserMFA action.

IAM users are not designed for end-user authentication at the scale required. While IAM allows MFA management, it does not provide the adaptive, risk-based authentication needed to enforce 2FA based on login context.

- ⊗ Configure AWS IAM Identity Center (AWS Single Sign-On) authentication for user authentication. Configure the permission sets to require multi-factor authentication (MFA).

AWS IAM Identity Center (successor to AWS SSO) can enforce MFA, but doesn't natively provide adaptive authentication based on location, IP address, or device inconsistencies without additional integration.

Overall Explanation:

Amazon Cognito user pools with risk-based adaptive authentication is the only solution that directly meets all requirements. It provides MFA, scales to millions of users, and allows dynamic enforcement of MFA based on the risk profile of each login attempt, considering factors like location, IP address, and device. Adaptive authentication reviews location and device information from users' sign-in requests and applies an automatic response to secure user accounts against suspicious activity.

 View Discussions

 Ask AI

Question 24

Not Attempted

Database



A company maintains an Amazon S3 data lake and needs a daily process to transform and load data into a data warehouse. The data warehouse requires massively parallel processing (MPP) capabilities. Data analysts will then use SQL commands to create and train machine learning (ML) models on the data. The solution should prioritize serverless AWS services where possible. Which solution best meets these requirements?

- ⊗ Run a daily Amazon EMR job to transform the data and load the data into Amazon Redshift. Use Amazon Redshift ML to create and train the ML models.

While EMR can transform data and Redshift can be used for ML, this option doesn't explicitly highlight the serverless aspect for EMR and may incur unnecessary overhead.

- ☐ Run a daily Amazon EMR job to transform the data and load the data into Amazon Aurora Serverless. Use Amazon Aurora ML to create and train the ML models.
- Aurora Serverless isn't primarily designed as a data warehouse for large-scale analytics with MPP. While Aurora ML exists, it's not the most suitable solution for the described scenario.
- ☒ Run a daily AWS Glue job to transform the data and load the data into Amazon Redshift Serverless. Use Amazon Redshift ML to create and train the ML models.
- This option utilizes serverless components (AWS Glue and Redshift Serverless) and effectively addresses the MPP requirement for the data warehouse. Redshift ML also meets the needs for creating and training ML models using SQL.
- ☐ Run a daily AWS Glue job to transform the data and load the data into Amazon Athena tables. Use Amazon Athena ML to create and train the ML models.
- While Glue and Athena are serverless, Athena isn't optimized as a data warehouse for large-scale analytics in the same way Redshift is. Although Athena can be used for ML, it's not the ideal choice given the specified requirements.

Overall Explanation:

The best solution is to use AWS Glue to transform the data and load it into Amazon Redshift Serverless. This combination leverages serverless services for both ETL and data warehousing, satisfying the MPP requirement and enabling data analysts to create and train ML models using SQL with Redshift ML. Redshift Serverless provides an on-demand, auto-scaling data warehouse that aligns with the company's need for a data warehouse that can handle large datasets and complex queries.

[View Discussions](#)[Ask AI](#)

Question 25

Not Attempted

[Compute](#)

A company runs containers in a Kubernetes environment in the company's local data center. The company wants to use Amazon Elastic Kubernetes Service (Amazon EKS) and other AWS managed services. Data must remain locally in the company's data center and cannot be stored in any remote site or cloud to maintain compliance. Which solution will meet these requirements?

- ☐ Deploy AWS Local Zones in the company's data center.

AWS Local Zones provide low-latency access to AWS services by placing AWS infrastructure closer to customers, but the data is still stored in an AWS-managed facility.

✗ Use an AWS Snowmobile in the company's data center.

AWS Snowmobile is used for large-scale data migrations to the cloud, not for running workloads or storing data locally.

✓ Install an AWS Outposts rack in the company's data center.

AWS Outposts allows you to run AWS services, including Amazon EKS, in your own data center while keeping data locally.

✗ Install an AWS Snowball Edge Storage Optimized node in the data center.

AWS Snowball Edge is a portable edge computing and data storage device, but it's not suitable for running Kubernetes or AWS-managed services like Amazon EKS.

Overall Explanation:

AWS Outposts is the service that allows you to run AWS infrastructure and services in your own data center, which is the correct solution to meet the requirements of keeping data local while using Amazon EKS. Local Zones, Snowmobile and Snowball edge do not fulfil this requirement.

 View Discussions

 Ask AI

Question 26

Not Attempted

Storage



A social media company's workloads collect and process data, storing it in on-premises NFS storage. The data store is struggling to scale to meet the company's growing business needs. The company wants to migrate the current data store to AWS in the most cost-effective way. Which solution should they implement?

✗ Set up an AWS Storage Gateway Volume Gateway. Use an Amazon S3 Lifecycle policy to transition the data to the appropriate storage class.

This option involves Volume Gateway, which presents cloud-backed storage volumes to on-premises applications. While it enables migration, it might not be the most cost-effective for NFS-based workloads specifically, as File Gateway is designed for this purpose.

✓ Set up an AWS Storage Gateway Amazon S3 File Gateway. Use an Amazon S3 Lifecycle policy to transition the data to the appropriate storage class.

This is the most suitable option. File Gateway allows storing files as objects in S3, leveraging S3's scalability and cost-effectiveness. The lifecycle policy further optimizes costs by transitioning data to cheaper storage tiers based on access patterns. It integrates with existing NFS workflows.

- ⊗ Use the Amazon Elastic File System (Amazon EFS) Standard-Infrequent Access (Standard-IA) storage class. Activate the infrequent access lifecycle policy.

While EFS is a suitable option for file storage in AWS, it requires changes to the existing application workflows, which are based on NFS. It also lacks direct integration for migrating from on-premises NFS in a cost-effective manner.

- ⊗ Use the Amazon Elastic File System (Amazon EFS) One Zone-Infrequent Access (One Zone-IA) storage class. Activate the infrequent access lifecycle policy.

Similar to the EFS Standard-IA option, this requires changes to existing application workflows and does not provide a streamlined migration path from on-premises NFS. Also, it only provides single-AZ availability.

Overall Explanation:

The most cost-effective solution is to set up an AWS Storage Gateway Amazon S3 File Gateway and use an Amazon S3 Lifecycle policy. This allows the company to migrate its on-premises NFS data store to AWS without significant changes to existing applications or workflows. AWS Storage Gateway provides seamless and secure integration between on-premises and AWS storage. Amazon S3 File Gateway provides a file interface to Amazon S3, with local caching for low-latency access. The lifecycle policy ensures that data is moved to cheaper storage tiers as it ages, further reducing costs.

[View Discussions](#)[Ask AI](#)

Question 27

Not Attempted

Serverless



A company uses high concurrency AWS Lambda functions to process a constantly increasing number of messages in a message queue during marketing events. The Lambda functions use CPU intensive code to process the messages. The company wants to reduce the compute costs and to maintain service latency for its customers. Which solution will meet these requirements?

- ☒ **Configure reserved concurrency for the Lambda functions. Decrease the memory allocated to the Lambda functions.**

Reserved concurrency guarantees the maximum number of concurrent instances but doesn't address cold starts, and decreasing memory can negatively impact CPU-intensive tasks.
- ☒ **Configure reserved concurrency for the Lambda functions. Increase the memory according to AWS Compute Optimizer recommendations.**

Reserved concurrency guarantees the maximum number of concurrent instances but doesn't address cold starts. Increasing memory based on Compute Optimizer recommendations can improve performance.
- ☒ **Configure provisioned concurrency for the Lambda functions. Decrease the memory allocated to the Lambda functions.**

Provisioned concurrency addresses cold starts but decreasing memory will negatively impact CPU-intensive tasks, hindering performance and potentially increasing costs due to longer execution times.
- ☒ **Configure provisioned concurrency for the Lambda functions. Increase the memory according to AWS Compute Optimizer recommendations.**

Provisioned concurrency ensures Lambda functions are initialized and ready to process incoming events, reducing cold start latency. Increasing memory, as recommended by AWS Compute Optimizer, can improve performance for CPU-intensive tasks, potentially reducing execution time and lowering costs.

Overall Explanation:

The best solution is to configure provisioned concurrency and increase the memory according to AWS Compute Optimizer recommendations. Provisioned concurrency addresses cold starts and maintains service latency, which is critical during marketing events with high message volume. Increasing memory allows the Lambda function to execute the CPU-intensive code more efficiently, potentially reducing execution time and thus cost. Reserved concurrency does not eliminate cold starts. Decreasing the memory allocated would also have negative impact on the performance.

[View Discussions](#)[Ask AI](#)

Question 28

Not Attempted

Security



A company runs its workloads on Amazon Elastic Container Service (Amazon ECS). The container images that the ECS task definition uses need to be scanned for Common Vulnerabilities and Exposures (CVEs). New container images that are created also need to be scanned. The solution must introduce minimal changes to existing workloads. Which solution will meet these requirements with the FEWEST changes to the workloads?

- ☒ Use Amazon Elastic Container Registry (Amazon ECR) as a private image repository to store the container images. Specify scan on push filters for the ECR basic scan.
Using ECR with scan-on-push filters provides CVE scanning with minimal changes to the workloads, as the images are already stored and retrieved from a container registry. Basic scan is sufficient for CVEs.
- ☐ Store the container images in an Amazon S3 bucket. Use Amazon Macie to scan the images. Use an S3 Event Notification to initiate a Macie scan for every event with an s3:ObjectCreated:Put event type.
Amazon Macie is not designed for scanning container images for CVEs. It focuses on data discovery and classification within S3 buckets.
- ☐ Deploy the workloads to Amazon Elastic Kubernetes Service (Amazon EKS). Use Amazon Elastic Container Registry (Amazon ECR) as a private image repository. Specify scan on push filters for the ECR enhanced scan.
Migrating to EKS introduces significant changes to the workloads and infrastructure. ECR would provide CVE scanning, but the switch to EKS is unnecessary and increases operational overhead.
- ☐ Store the container images in an Amazon S3 bucket that has versioning enabled. Configure an S3 Event Notification for s3:ObjectCreated:* events to invoke an AWS Lambda function. Configure the Lambda function to initiate an Amazon Inspector scan.
This solution involves a more complex setup using S3, Lambda, and Inspector, requiring additional configuration and maintenance. It is less straightforward and involves more steps compared to using ECR scan-on-push features.

Overall Explanation:

Using Amazon Elastic Container Registry (ECR) as a private image repository with scan-on-push filters for the ECR basic scan provides the simplest and most direct solution for scanning container images for CVEs. It requires the fewest changes to the existing ECS workloads and leverages built-in functionality. Option B uses Macie which is not for CVE scans. Option C involves a full migration of ECS to EKS which is a very large change to the workloads. Option D, while it might work is far more complicated than option A.

[View Discussions](#)[Ask AI](#)

Question 29

Not Attempted

Serverless



A company uses an AWS Batch job to run its end-of-day sales process. The company needs a serverless solution that will invoke a third-party reporting application when the AWS Batch job is successful. The reporting application has an HTTP API interface that uses username and password authentication. Which solution will meet these requirements?

- ☒ Configure an Amazon EventBridge rule to match incoming AWS Batch job SUCCEEDED events. Configure the third-party API as an EventBridge API destination with a username and password. Set the API destination as the EventBridge rule target.

This option is the most direct and serverless approach. EventBridge detects successful AWS Batch job completion and triggers actions. By configuring the third-party API as an API destination with authentication credentials, EventBridge directly invokes the application without extra services, minimizing complexity and operational overhead. API Destinations specifically support authentication.

- ☐ Configure Amazon EventBridge Scheduler to match incoming AWS Batch job SUCCEEDED events. Configure an AWS Lambda function to invoke the third-party API by using a username and password. Set the Lambda function as the EventBridge rule target.

While this option could work, EventBridge Scheduler is not needed. EventBridge rules can already match Batch job events. Adding a Lambda function introduces unnecessary complexity and cost.

- ☐ Configure an AWS Batch job to publish job SUCCEEDED events to an Amazon API Gateway REST API. Configure an HTTP proxy integration on the API Gateway REST API to invoke the third-party API by using a username and password.

An HTTP proxy integration on API Gateway cannot directly handle authentication. Separate management of username and password authentication is still required, adding complexity.

- ☐ Configure an AWS Batch job to publish job SUCCEEDED events to an Amazon API Gateway REST API. Configure a proxy integration on the API Gateway REST API to an AWS Lambda function. Configure the Lambda function to invoke the third-party API by using a username and password.

This is more complex than other solutions. Introducing API Gateway and Lambda adds unnecessary layers when EventBridge API Destinations can handle the task directly.

Overall Explanation:

The best solution is to use Amazon EventBridge to detect the successful AWS Batch job completion and trigger the third-party API using API Destinations. EventBridge API Destinations can handle the HTTP API calls and manage authentication directly, providing a serverless and streamlined approach. Using the EventBridge scheduler or adding Lambda functions or API Gateway integrations adds unnecessary complexity.

[View Discussions](#)[Ask AI](#)

Question 30

Not Attempted

[Compute](#)

A solutions architect is designing a cloud architecture for a new application deployed on AWS. The application must process jobs in parallel, scaling application nodes up or down based on the number of jobs awaiting processing. The processor application is stateless. The architect needs to ensure the application is loosely coupled and that job items are durably stored. Which design meets these requirements?

- ☐ Create an Amazon SNS topic to send jobs. Create an AMI for the processor application. Create a launch configuration using the AMI. Create an Auto Scaling group using the launch configuration, scaling based on CPU usage.

SNS is a publish/subscribe service and doesn't provide the durability needed for job storage. CPU usage may not accurately reflect the number of jobs waiting.

- ☐ Create an Amazon SQS queue to hold jobs. Create an AMI for the processor application. Create a launch configuration using the AMI. Create an Auto Scaling group using the launch configuration, scaling based on network usage.

While SQS provides durability, network usage isn't the best metric for scaling based on the number of jobs. Launch configurations are older technology.

- ☒ Create an Amazon SQS queue to hold jobs. Create an AMI for the processor application. Create a launch template using the AMI. Create an Auto Scaling group using the launch template, scaling based on the number of items in the SQS queue.

This design uses SQS for durable storage and loose coupling. The launch template and Auto Scaling group allow dynamic scaling based on the number of jobs in the queue, ensuring parallel processing.

- ☐ Create an Amazon SNS topic to send jobs. Create an AMI for the processor application. Create a launch template using the AMI. Create an Auto Scaling group using the launch template, scaling based on the number of messages published to the SNS topic.

SNS doesn't guarantee durability. Scaling based on messages published to SNS doesn't directly correlate to the number of unprocessed jobs.

Overall Explanation:

The best solution uses SQS for durable and loosely coupled job storage. A launch template and Auto Scaling group enable dynamic scaling of application nodes based on the number of items in the SQS queue, facilitating parallel processing of jobs. SNS does not provide the necessary durability, and scaling based on CPU, network usage or published messages does not accurately reflect the number of jobs requiring processing. SQS also supports durable storage of job items and launch templates are preferred over launch configurations.

[View Discussions](#)[Ask AI](#)

Question 31

Not Attempted

Networking



A company needs to access data from a vendor's Amazon RDS for MySQL database. The vendor's database resides in their own AWS account. The company's VPC lacks an internet gateway, AWS Direct Connect, and AWS Site-to-Site VPN. How can the company securely access the vendor's database, ensuring no public internet exposure?

- ☒ Instruct the vendor to establish an AWS Hosted Connection Direct Connect Program and configure VPC peering between the company's and vendor's VPCs.
This option requires Direct Connect, which is not a feasible solution given the constraints mentioned in the question.
- ☒ Set up a client VPN connection between the company's VPC and the vendor's VPC and then configure VPC peering between the two VPCs.
VPC peering does not circumvent the need for an internet gateway or other connections to outside the VPC.
- ☒ Instruct the vendor to create a Network Load Balancer (NLB) in front of the Amazon RDS for MySQL database and use AWS PrivateLink to establish a connection between the company's VPC and the vendor's VPC.
AWS PrivateLink allows private connectivity to services hosted in another VPC without exposing them to the public internet. The NLB serves as the integration point for PrivateLink, routing traffic to the RDS database.

- ☐ Employ AWS Transit Gateway to integrate the company's VPC and the vendor's VPC, then use VPC peering to connect the VPCs.

Transit Gateway with VPC peering is overly complex and unnecessary for this scenario. PrivateLink offers a simpler and more secure solution.

Overall Explanation:

The most appropriate solution is to leverage AWS PrivateLink. This allows the vendor to create an NLB in front of their RDS database and expose it as a PrivateLink service. The company can then create a VPC endpoint to privately connect to the vendor's service without requiring an internet gateway, Direct Connect, or VPN. PrivateLink provides secure and private access to services across accounts and VPCs. VPC peering, Direct Connect, and VPN are not required.

 View Discussions

 Ask AI

Question 32

Not Attempted

Monitoring and Observability



A company wants to use Amazon Managed Grafana to visualize data from its Amazon RDS database as a single data source. The company's security policy mandates that the solution must not expose the data over the internet. Which of the following solutions meets these requirements?

- ☐ Create an Amazon Managed Grafana workspace without a VPC. Create a public endpoint for the RDS database. Configure the public endpoint as a data source in Amazon Managed Grafana.
- This solution uses a public endpoint, which violates the requirement that the data should not be exposed over the internet.
- ☒ Create an Amazon Managed Grafana workspace in a VPC. Create a private endpoint for the RDS database. Configure the private endpoint as a data source in Amazon Managed Grafana.
- This option uses a VPC for Grafana and a private endpoint for the database, ensuring that the data traffic remains within the AWS network and is not exposed to the internet.
- ☐ Create an Amazon Managed Grafana workspace without a VPC. Create an AWS PrivateLink endpoint to establish a connection between Amazon Managed Grafana and

Amazon RDS. Set up Amazon RDS as a data source in Amazon Managed Grafana.

Amazon Managed Grafana must be within a VPC to utilize a PrivateLink endpoint to communicate with RDS. Without a VPC, this is not possible.

- ⊗ Create an Amazon Managed Grafana workspace in a VPC. Create a public endpoint for the RDS database. Configure the public endpoint as a data source in Amazon Managed Grafana.

This option uses a public endpoint, which violates the requirement that the data should not be exposed over the internet, despite Grafana being within a VPC.

Overall Explanation:

The correct answer is to create an Amazon Managed Grafana workspace in a VPC and use a private endpoint for the RDS database. This ensures that all traffic between Grafana and RDS remains within the AWS network and is not exposed to the public internet, meeting the company's security requirements.

 View Discussions

 Ask AI

Question 33

Not Attempted

Storage



A company hosts a data lake on Amazon S3. The data lake ingests data in Apache Parquet format from various data sources. The company uses multiple transformation steps to prepare the ingested data. The steps include filtering of anomalies, normalizing of data to standard date and time values, and generation of aggregates for analyses. The company must store the transformed data in S3 buckets that data analysts access. The company needs a prebuilt solution for data transformation that does not require code. The solution must provide data lineage and data profiling. The company needs to share the data transformation steps with employees throughout the company. Which solution will meet these requirements?

- ⊗ Configure an AWS Glue Studio visual canvas to transform the data. Share the transformation steps with employees by using AWS Glue jobs.

AWS Glue Studio requires more coding knowledge than a no-code solution. It also doesn't natively provide sharing of visual canvas transformation steps.

- ⊗ Configure Amazon EMR Serverless to transform the data. Share the transformation steps with employees by using EMR Serverless jobs.

Amazon EMR Serverless requires coding and is not a no-code solution. It doesn't provide prebuilt data profiling or lineage capabilities.

- ✔ Configure AWS Glue DataBrew to transform the data. Share the transformation steps with employees by using DataBrew recipes.

AWS Glue DataBrew is a no-code data preparation tool that includes features for data lineage, data profiling, and easily sharable recipes for data transformation.

- ✘ Create Amazon Athena tables for the data. Write Athena SQL queries to transform the data. Share the Athena SQL queries with employees.

While Athena can perform data transformation using SQL, writing SQL queries is considered coding. Athena also lacks built-in data lineage and profiling features.

Overall Explanation:

AWS Glue DataBrew provides a visual, no-code environment for data transformation. It supports data lineage, data profiling, and the sharing of transformation steps via DataBrew recipes. This meets all of the company's requirements. Glue Studio requires more coding knowledge than a no-code solution. Amazon EMR Serverless requires coding and Athena also requires coding skills.

[View Discussions](#)[Ask AI](#)

Question 34

Not Attempted

Networking



A solutions architect runs a web application on multiple Amazon EC2 instances in individual target groups behind an Application Load Balancer (ALB). Users access the application through a public website. The architect needs to provide engineers with access to a development version of the website, allowing them to test new features on a specific development EC2 instance. An Amazon Route 53 hosted zone should enable this access. The solution must automatically route to the development instance, even if the instance is replaced. Which solution will meet these requirements?

- ✔ Create an A Record for the development website with the value set to the ALB. Create a listener rule on the ALB that forwards requests for the development website to the target group containing the development instance.

This solution allows the ALB to route traffic based on the host header to the development instance's target group. When the development instance is replaced, the target group will be automatically

updated with the new instance, requiring no DNS changes.

- ⊗ Recreate the development instance with a public IP address. Create an A Record for the development website with the value set to the public IP address of the development instance.

If the development instance is replaced, its public IP address might change, requiring manual DNS updates in Route 53. This solution does not provide automatic routing after instance replacement.

- ⊗ Create an A Record for the development website with the value set to the ALB. Create a listener rule on the ALB to redirect requests for the development website to the public IP address of the development instance.

ALB listener rules are not intended to directly redirect traffic to the public IP addresses of EC2 instances. Redirecting to a public IP bypasses the benefits of using a target group for health checks and automated replacement.

- ⊗ Place all instances in the same target group. Create an A Record for the development website. Set the value to the ALB. Create a listener rule on the ALB that forwards requests for the development website to the target group.

Placing all instances in the same target group does not isolate the development instance. Requests intended for the development website might be incorrectly routed to production instances, and it will be difficult to isolate the Dev Environment.

Overall Explanation:

The best solution is to create an A Record pointing to the ALB and create a listener rule on the ALB to forward requests for the development website to the target group containing the development instance. This setup ensures that traffic is correctly routed to the development instance and that routing is automatically updated when the instance is replaced.

[View Discussions](#)[Ask AI](#)

Question 35

Not Attempted

[Compute](#)

A company runs a container application on a Kubernetes cluster in the company's data center. The application uses Advanced Message Queuing Protocol (AMQP) to communicate with a message queue. The data center cannot scale fast enough to meet the company's expanding

business needs. The company wants to migrate the workloads to AWS. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Migrate the container application to Amazon Elastic Container Service (Amazon ECS). Use Amazon Simple Queue Service (Amazon SQS) to retrieve the messages.

While ECS is a good container orchestration service and SQS is a managed queueing service, SQS does not support the AMQP protocol. Therefore, this option requires application code changes.

- ✓ Migrate the container application to Amazon Elastic Kubernetes Service (Amazon EKS). Use Amazon MQ to retrieve the messages.

This option allows the company to use a familiar container orchestration platform (Kubernetes) via EKS. Amazon MQ provides compatibility with AMQP, minimizing code changes and operational overhead.

- ⊗ Use highly available Amazon EC2 instances to run the application. Use Amazon MQ to retrieve the messages.

While this solution works, it requires more manual configuration and management of EC2 instances, increasing operational overhead compared to a managed container service.

- ⊗ Use AWS Lambda functions to run the application. Use Amazon Simple Queue Service (Amazon SQS) to retrieve the messages.

This option would likely require a significant rewrite of the container application to fit the Lambda execution model. Additionally, SQS doesn't support AMQP, adding further complexity.

Overall Explanation:

The best option is to migrate the application to Amazon EKS and use Amazon MQ. EKS allows the company to continue using Kubernetes, and Amazon MQ provides compatibility with the AMQP protocol, which is required by the application. This approach minimizes code changes and operational overhead. While ECS + SQS offers less overhead in general, the lack of AMQP support in SQS makes it unsuitable. Using EC2 directly increases operational overhead compared to managed services. Lambda introduces significant application rewrite requirements and doesn't address AMQP.

 View Discussions

 Ask AI

Question 36

Not Attempted

Networking



An online gaming company hosts its platform on Amazon EC2 instances behind Network Load Balancers (NLBs) across multiple AWS Regions. The NLBs can route requests to targets over the internet. The company wants to improve the customer playing experience by reducing end-to-end load time for its global customer base. Which solution will meet these requirements?

- ☐ Create Application Load Balancers (ALBs) in each Region to replace the existing NLBs. Register the existing EC2 instances as targets for the ALBs in each Region.

ALBs are typically used for HTTP/HTTPS traffic and don't provide the same level of performance improvement for gaming traffic as Global Accelerator. Also, this option involves significant changes to the existing infrastructure.

- ☐ Configure Amazon Route 53 to route equally weighted traffic to the NLBs in each Region.

Equally weighted traffic distribution does not optimize for latency and proximity to the user. It will not significantly reduce end-to-end load time for the global customer base.

- ☐ Create additional NLBs and EC2 instances in other Regions where the company has large customer bases.

While this can help, it doesn't guarantee optimal routing and requires more infrastructure management. It also doesn't address the inherent latency of internet routing.

- ☒ Create a standard accelerator in AWS Global Accelerator. Configure the existing NLBs as target endpoints.

AWS Global Accelerator uses the AWS global network to direct traffic to the closest healthy endpoint, minimizing latency and improving the user experience for a global customer base. Because the infrastructure currently uses NLBs, Global Accelerator is an optimal solution.

Overall Explanation:

AWS Global Accelerator is designed to improve the performance of applications by directing traffic to the optimal endpoint based on user location and network conditions. By creating a standard accelerator and configuring the existing NLBs as target endpoints, the gaming company can leverage the AWS global network to reduce latency and improve the playing experience for its global customer base. Option A would be better if it implemented Cloudfront as well, but it doesn't. Route 53 routing doesn't optimize for latency. Creating additional NLBs does not guarantee optimal routing. Therefore, option D is the best answer.

[View Discussions](#)[Ask AI](#)

Question 37

Not Attempted

Serverless



A company is migrating to the AWS Cloud. Currently, an on-premises application uses SFTP to collect financial data from multiple vendors. The company has created a new cloud application that utilizes Amazon S3 APIs for file uploads from vendors. However, some vendors use legacy applications that do not support S3 APIs and want to continue using SFTP. The company wants to use managed services to accommodate these vendors with minimal operational overhead. Which solution meets these requirements with the LEAST operational overhead?

- ☐ Create an AWS Database Migration Service (AWS DMS) instance to replicate data from the storage of the vendors that use legacy applications to Amazon S3. Provide the vendors with the credentials to access the AWS DMS instance.

AWS DMS is designed for database migrations, not file-based SFTP uploads.

- ☒ Create an AWS Transfer Family endpoint for vendors that use legacy applications.

AWS Transfer Family is a fully managed service that directly supports SFTP, FTPS, and FTP protocols, providing a managed SFTP endpoint for vendors to upload data directly to Amazon S3, minimizing operational overhead.

- ☐ Configure an Amazon EC2 instance to run an SFTP server. Instruct the vendors that use legacy applications to use the SFTP server to upload data.

This solution requires managing and maintaining an EC2 instance, increasing operational overhead.

- ☐ Configure an Amazon S3 File Gateway for vendors that use legacy applications to upload files to an SMB file share.

S3 File Gateway supports SMB and NFS, not SFTP.

Overall Explanation:

AWS Transfer Family is a fully managed service that allows you to set up SFTP endpoints for accessing Amazon S3 storage. By creating an AWS Transfer Family endpoint, the company can provide vendors with a familiar SFTP interface to upload data directly to Amazon S3 without requiring changes to their legacy applications. This eliminates the need for the company to manage additional infrastructure such as EC2 instances or file gateways, reducing operational overhead.

[View Discussions](#)[Ask AI](#)

Question 38

Not Attempted

Serverless



A marketing team is preparing a campaign for an upcoming multi-sport event. They possess a collection of news reports in PDF format spanning the past five years. The team needs a solution to extract key insights and analyze the sentiment expressed within these news reports. The solution must leverage Amazon Textract to process the PDF documents. Which solution meets these requirements with the LEAST operational overhead?

- ☐ Provide the extracted insights to Amazon Athena for analysis. Store the extracted insights and analysis in an Amazon S3 bucket.
Using Athena would require creating and managing tables and queries, increasing operational overhead. It's also not ideally suited for sentiment analysis.
- ☐ Store the extracted insights in an Amazon DynamoDB table. Use Amazon SageMaker to build a sentiment model.
Building and managing a SageMaker sentiment model introduces significant operational complexity and overhead. DynamoDB is also not optimized for this type of analytical workload.
- ☒ Provide the extracted insights to Amazon Comprehend for analysis. Save the analysis to an Amazon S3 bucket.
Amazon Comprehend provides built-in sentiment analysis capabilities with minimal operational overhead. It integrates well with Textract and S3, providing a streamlined solution.
- ☐ Store the extracted insights in an Amazon S3 bucket. Use Amazon QuickSight to visualize and analyze the data.
QuickSight is primarily for visualization and requires additional steps to perform sentiment analysis, increasing operational overhead. It also does not provide built in sentiment analysis.

Overall Explanation:

The most efficient solution involves using Amazon Textract to extract text from the news reports and then leveraging Amazon Comprehend's built-in sentiment analysis capabilities. The results can then be stored in an S3 bucket. This approach minimizes operational overhead by utilizing managed services specifically designed for these tasks.

[View Discussions](#)[Ask AI](#)

Question 39

Not Attempted

Serverless



A company's application runs on Amazon EC2 instances that are in multiple Availability Zones. The application needs to ingest real-time data from third-party applications. The company needs a data ingestion solution that places the ingested raw data in an Amazon S3 bucket. Which solution will meet these requirements?

- ☒ Create Amazon Kinesis Data Streams for data ingestion. Create Amazon Kinesis Data Firehose delivery streams to consume the Kinesis Data Streams. Specify the S3 bucket as the destination of the delivery streams.

This solution utilizes Kinesis Data Streams for real-time data ingestion, and Kinesis Data Firehose for delivering the data to an S3 bucket. This fulfills the requirements of real-time ingestion and S3 storage.

- ☐ Create database migration tasks in AWS Database Migration Service (AWS DMS). Specify replication instances of the EC2 instances as the source endpoints. Specify the S3 bucket as the target endpoint. Set the migration type to migrate existing data and replicate ongoing changes.

AWS DMS is primarily for database migrations, not real-time data ingestion from applications. It's not suitable for handling the real-time data ingestion requirement.

- ☐ Create and configure AWS DataSync agents on the EC2 instances. Configure DataSync tasks to transfer data from the EC2 instances to the S3 bucket.

AWS DataSync is designed for periodic data transfers, not real-time ingestion. It's not suitable for handling the real-time data ingestion requirement.

- ☐ Create an AWS Direct Connect connection to the application for data ingestion. Create Amazon Kinesis Data Firehose delivery streams to consume direct PUT operations from the application. Specify the S3 bucket as the destination of the delivery streams.

AWS Direct Connect is about creating dedicated network connections; it doesn't inherently solve the data ingestion problem itself. Also, Kinesis Firehose is generally used for ingestion from Kinesis Streams, not directly from applications via PUT operations.

Overall Explanation:

Kinesis Data Streams and Kinesis Data Firehose is the most suitable solution for ingesting real-time data from third-party applications and storing it in an S3 bucket. Kinesis Data

Streams handles the real-time ingestion, while Kinesis Data Firehose efficiently delivers the data to the specified S3 destination.

[View Discussions](#)[Ask AI](#)

Question 40

Not Attempted

Database



A company's application receives data from multiple sources, with data sizes varying up to 700 KB and expected to increase. The company has chosen Amazon DynamoDB as its primary database. A solutions architect needs a solution that handles these potentially large data sizes operationally efficiently. Which of the following approaches is the MOST operationally efficient for storing data exceeding DynamoDB's item size limits?

- ☐ Create an AWS Lambda function to filter data exceeding DynamoDB item size limits and store the larger data in an Amazon DocumentDB (with MongoDB compatibility) database.

Using DocumentDB introduces another database technology into the stack, increasing operational complexity and management overhead. Filtering data might lead to data loss.

- ☒ Store the large data as objects in an Amazon S3 bucket. In a DynamoDB table, create an item with an attribute pointing to the S3 URL of the data.

This approach decouples large data storage from DynamoDB, leveraging S3's scalability and cost-effectiveness for large objects. DynamoDB stores metadata and pointers to the actual data.

- ☐ Split all incoming large data into a collection of items that share the same partition key. Write the data to a DynamoDB table in a single operation using the BatchWriteItem API operation.

Splitting data into multiple items adds complexity to both writing and reading the data, and BatchWriteItem has its own limitations.

- ☐ Create an AWS Lambda function that uses gzip compression to compress large objects as they are written to a DynamoDB table.

Compression reduces size, but doesn't guarantee the data will fit within the DynamoDB's 400 KB limit, especially with increasing data sizes. Also, compression and decompression add computational overhead.

Overall Explanation:

Storing large data in Amazon S3 and referencing it with a URL in DynamoDB offers the most operationally efficient solution. It leverages S3's ability to store large objects cost-effectively, while DynamoDB manages metadata and pointers to the S3 objects. This approach avoids the complexities of using another database or splitting data into multiple DynamoDB items and is in line with AWS recommended practices. Compression may help, but it's not guaranteed to keep the data under the DynamoDB size limit as data sizes grow.

[View Discussions](#)[Ask AI](#)

Question 41

Not Attempted

Security



A company hosts its web applications in the AWS Cloud. The company configures Elastic Load Balancers to use certificates that are imported into AWS Certificate Manager (ACM). The company's security team must be notified 30 days before the expiration of each certificate. What should a solutions architect recommend to meet this requirement?

- ☐ Add a rule in ACM to publish a custom message to an Amazon Simple Notification Service (Amazon SNS) topic every day, beginning 30 days before any certificate will expire.
ACM does not directly support custom rules for sending notifications directly to SNS based on specific conditions like certificate expiration. EventBridge is the mechanism for ACM notifications.
- ☒ Create an AWS Config rule that checks for certificates that will expire within 30 days. Configure Amazon EventBridge (Amazon CloudWatch Events) to invoke a custom alert by way of Amazon Simple Notification Service (Amazon SNS) when AWS Config reports a noncompliant resource.
AWS Config can check for expiring certificates using the `acm-certificate-expiration-check` managed rule. When a certificate is found to be non-compliant (expiring within the specified timeframe), EventBridge can trigger an SNS notification.
- ☐ Use AWS Trusted Advisor to check for certificates that will expire within 30 days. Create an Amazon CloudWatch alarm that is based on Trusted Advisor metrics for check status changes. Configure the alarm to send a custom alert by way of Amazon Simple Notification Service (Amazon SNS).
Trusted Advisor does not monitor certificate expiration and is not the appropriate tool for this task.

- ⊗ Create an Amazon EventBridge (Amazon CloudWatch Events) rule to detect any certificates that will expire within 30 days. Configure the rule to invoke an AWS Lambda function. Configure the Lambda function to send a custom alert by way of Amazon Simple Notification Service (Amazon SNS).

While EventBridge can detect certificate expiration events, it is more complex to create custom logic within EventBridge or Lambda to look ahead 30 days. AWS Config provides a managed rule specifically for this purpose, making Option B more efficient.

Overall Explanation:

The best solution is to use AWS Config with the ``acm-certificate-expiration-check`` managed rule. AWS Config continuously monitors the compliance of AWS resources. EventBridge will then trigger an SNS topic, which allows you to notify the security team when the certificates expire within 30 days. This approach simplifies the notification process, making it more reliable and easier to manage.

 View Discussions

 Ask AI

Question 42

Not Attempted

Serverless



A company is migrating a legacy application from an on-premises data center to AWS. The application relies on hundreds of cron jobs that run between 1 and 20 minutes on different recurring schedules throughout the day. The company wants a solution to schedule and run the cron jobs on AWS with minimal refactoring. The solution must support running the cron jobs in response to an event in the future. Which solution will meet these requirements?

- ⊗ Create a container image for the cron jobs. Use Amazon EventBridge Scheduler to create a recurring schedule. Run the cron job tasks as AWS Lambda functions.
- This solution is incorrect because AWS Lambda functions have a maximum execution time limit of 15 minutes. The cron jobs run for up to 20 minutes, which exceeds the Lambda limit.
- ⊗ Create a container image for the cron jobs. Use AWS Batch on Amazon Elastic Container Service (Amazon ECS) with a scheduling policy to run the cron jobs.
- This solution is a possible solution, but not optimal because you would have to run highly-available virtual machines or containers waiting for the event to happen. This solution has higher operational overhead compared to Fargate.

- ✓ Create a container image for the cron jobs. Use Amazon EventBridge Scheduler to create a recurring schedule. Run the cron job tasks on AWS Fargate.

This solution is correct because Amazon EventBridge Scheduler can create recurring schedules to trigger tasks. AWS Fargate allows running containerized applications without managing the underlying infrastructure and can execute tasks longer than 15 minutes. Fargate allows you to pay only for the resources consumed during the execution of the cron jobs.

- ✗ Create a container image for the cron jobs. Create a workflow in AWS Step Functions that uses a Wait state to run the cron jobs at a specified time. Use the RunTask action to run the cron job tasks on AWS Fargate.

This solution is incorrect because it introduces unnecessary complexity by using Step Functions. While Step Functions can schedule and execute tasks, EventBridge Scheduler is specifically designed for scheduling tasks and integrates directly with Fargate.

Overall Explanation:

The best solution is to use Amazon EventBridge Scheduler to create the recurring schedules and run the cron job tasks on AWS Fargate. EventBridge Scheduler is designed for scheduling and supports running tasks in response to events. AWS Fargate is a serverless compute engine for containers that allows you to run the cron jobs without managing the underlying infrastructure. Since Lambda has a 15-minute limit and the jobs can run for 20 minutes, Lambda cannot be used.

 View Discussions

 Ask AI

Question 43

Not Attempted

Networking



A company uses Salesforce and needs to load existing data and ongoing data changes from Salesforce to Amazon Redshift for analysis. The company has a strict requirement to avoid data transfer over the public internet. Which solution will meet these requirements with the LEAST development effort?

- ✗ Establish a VPN connection from the VPC to Salesforce. Use AWS Glue DataBrew to transfer data.

A VPN connection still utilizes the public internet, which violates the requirement. Additionally, AWS Glue DataBrew is not the most suitable choice for ongoing data changes.

- ⊗ Establish an AWS Direct Connect connection from the VPC to Salesforce. Use AWS Glue DataBrew to transfer data.

While AWS Direct Connect avoids the public internet, it involves significant setup effort and cost. Also, DataBrew is not optimal for loading ongoing data changes.

- ✔ Create an AWS PrivateLink connection in the VPC to Salesforce. Use Amazon AppFlow to transfer data.

AWS PrivateLink provides private connectivity, avoiding the public internet. Amazon AppFlow is a fully managed service specifically designed for integration between SaaS applications like Salesforce and AWS services like Amazon Redshift, minimizing development effort.

- ⊗ Create a VPC peering connection to Salesforce. Use Amazon AppFlow to transfer data.

VPC peering is for connecting two AWS VPCs. Salesforce is external to AWS and does not have a VPC, therefore VPC peering cannot be used to connect to Salesforce.

Overall Explanation:

The best solution is to create an AWS PrivateLink connection in the VPC to Salesforce and use Amazon AppFlow to transfer data. PrivateLink ensures data doesn't traverse the public internet, and AppFlow is a managed service optimized for Salesforce integration, minimizing development effort. AWS PrivateLink enables private connectivity between VPCs and supported AWS services. Amazon AppFlow simplifies data transfer between SaaS applications (like Salesforce) and AWS services (like Amazon Redshift).

 View Discussions

 Ask AI

Question 44

Not Attempted

Storage



A company recently migrated its application to AWS. The application runs on Amazon EC2 Linux instances in an Auto Scaling group across multiple Availability Zones. The application stores data in an Amazon Elastic File System (Amazon EFS) file system that uses EFS Standard-Infrequent Access storage. The application indexes the company's files. The index is stored in an Amazon RDS database. The company needs to optimize storage costs with some application and services changes. Which solution will meet these requirements MOST cost-effectively?

- ✔ Create an Amazon S3 bucket that uses an Intelligent-Tiering lifecycle policy. Copy all files to the S3 bucket. Update the application to use Amazon S3 API to store and

retrieve files.

This is the most cost-effective solution. S3 Intelligent-Tiering automatically moves data between frequent and infrequent access tiers based on usage patterns, optimizing storage costs. It also allows for application changes and uses the S3 API which provides scalability and durability.

- ⊗ Deploy Amazon FSx for Windows File Server file shares. Update the application to use CIFS protocol to store and retrieve files.

FSx for Windows File Server is not cost-effective and is designed for Windows-based applications. Switching to CIFS protocol is not beneficial in this scenario.

- ⊗ Deploy Amazon FSx for OpenZFS file system shares. Update the application to use the new mount point to store and retrieve files.

FSx for OpenZFS can be expensive and optimized for High-Performance Data analysis. It is not suitable for cost optimization in this scenario.

- ⊗ Create an Amazon S3 bucket that uses S3 Glacier Flexible Retrieval. Copy all files to the S3 bucket. Update the application to use Amazon S3 API to store and retrieve files as standard retrievals.

S3 Glacier Flexible Retrieval has longer retrieval times (up to a few hours), which makes it unsuitable for an application that requires faster access to the files. EFS Infrequent access provides millisecond retrieval time, and Glacier is not a like-for-like replacement.

Overall Explanation:

The most cost-effective solution is to use Amazon S3 Intelligent-Tiering. This allows for automatic data tiering based on access patterns, reducing storage costs. Other options are either more expensive or have unacceptable retrieval times. Switching to S3 also requires application changes, which the company is open to, making this a feasible solution.

Options B and C are more expensive. Option D has unacceptable retrieval times, rendering the application unusable.

 View Discussions

 Ask AI

Question 45

Not Attempted

Networking



A robotics company is developing a surgical robot using advanced sensors, cameras, and AI. They need a public load balancer in AWS to ensure seamless and encrypted communication

with backend services, capable of routing traffic based on query strings. Which solution best meets these requirements?

- ☐ Use a Network Load Balancer with a certificate attached from AWS Certificate Manager (ACM). Use query parameter-based routing.
Network Load Balancers do not support query parameter-based routing.
- ☐ Use a Gateway Load Balancer. Import a generated certificate in AWS Identity and Access Management (IAM). Attach the certificate to the load balancer. Use HTTP path-based routing.
Gateway Load Balancers do not support query parameter-based routing and importing certificates into IAM for use with load balancers is not the standard practice; ACM is preferred.
- ☒ Use an Application Load Balancer with a certificate attached from AWS Certificate Manager (ACM). Use query parameter-based routing.
Application Load Balancers support query parameter-based routing and using ACM for certificate management. This also provides encryption.
- ☐ Use a Network Load Balancer. Import a generated certificate in AWS Identity and Access Management (IAM). Attach the certificate to the load balancer. Use query parameter-based routing.
Network Load Balancers do not support query parameter-based routing. Certificates are typically managed via ACM, not IAM.

Overall Explanation:

The Application Load Balancer (ALB) is the correct choice because it supports query parameter-based routing, which is required by the company. It also allows the attachment of a certificate from AWS Certificate Manager (ACM) for encrypting traffic. Network Load Balancers and Gateway Load Balancers do not support query parameter-based routing.

[View Discussions](#)[Ask AI](#)

Question 46

Not Attempted

Database



A company's application currently runs on a single Amazon EC2 instance, with a MySQL database hosted on the same instance. To handle increased traffic and ensure high availability and automatic scalability, what solution should the company implement?

- ⊗ Deploy the application to EC2 instances that run in an Auto Scaling group behind an Application Load Balancer. Create an Amazon Redshift cluster that has multiple MySQL-compatible nodes.

This option uses Auto Scaling and Application Load Balancer for application scaling but uses Amazon Redshift, which is a data warehousing service, and is not suitable for replacing a MySQL database.

- ⊗ Deploy the application to EC2 instances that are configured as a target group behind an Application Load Balancer. Create an Amazon RDS for MySQL cluster that has multiple instances.

This option uses an Application Load Balancer and target groups, which do not inherently provide automatic scaling. RDS for MySQL provides a scalable database solution but the lack of Auto Scaling for the application instances is a drawback.

- ✓ Deploy the application to EC2 instances that run in an Auto Scaling group behind an Application Load Balancer. Create an Amazon Aurora Serverless MySQL cluster for the database layer.

This option provides a highly available and automatically scalable solution. Auto Scaling ensures the application scales based on traffic, the Application Load Balancer distributes the traffic, and Aurora Serverless provides a scalable and managed MySQL-compatible database.

- ⊗ Deploy the application to EC2 instances that are configured as a target group behind an Application Load Balancer. Create an Amazon ElastiCache for Redis cluster that uses the MySQL connector.

This option uses ElastiCache for Redis, which is an in-memory data store, and is not suitable for replacing a MySQL database. Target groups also don't provide automatic scaling.

Overall Explanation:

The best solution for high availability and automatic scalability is to deploy the application to an Auto Scaling group behind an Application Load Balancer and use Amazon Aurora Serverless MySQL for the database layer. This ensures the application scales based on demand and the database is also automatically scalable. Redshift and ElastiCache are not direct replacements for MySQL in this scenario. Target Groups don't provide autoscaling. Therefore, option C is the correct answer.

 View Discussions

 Ask AI

Question 47

Not Attempted

Storage



A company is planning to migrate data to an Amazon S3 bucket. The data must be encrypted at rest within the S3 bucket. The encryption key must be automatically rotated every year. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ **✗** Migrate the data to the S3 bucket. Use server-side encryption with Amazon S3 managed keys (SSE-S3). Use the built-in key rotation behavior of SSE-S3 encryption keys.

SSE-S3 encryption keys are rotated regularly, but AWS does not guarantee a specific rotation period like every year. Also, encryption should be enabled **before** migrating the data. Thus, this option does not strictly meet the requirements.
- ☒ **✓** Create an AWS Key Management Service (AWS KMS) customer managed key. Enable automatic key rotation. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Migrate the data to the S3 bucket.

This solution correctly utilizes a KMS customer managed key with automatic key rotation enabled, ensuring encryption at rest and automatic yearly key rotation. This involves the least operational overhead as the rotation is handled automatically by KMS.
- ☐ **✗** Create an AWS Key Management Service (AWS KMS) customer managed key. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Migrate the data to the S3 bucket. Manually rotate the KMS key every year.

While using a KMS customer managed key will allow encryption at rest, manually rotating the key every year incurs operational overhead. The question specifies the solution with the LEAST operational overhead, so this is incorrect.
- ☐ **✗** Use customer key material to encrypt the data. Migrate the data to the S3 bucket. Create an AWS Key Management Service (AWS KMS) key without key material. Import the customer key material into the KMS key. Enable automatic key rotation.

Importing customer key material introduces more complexity and operational overhead than necessary. Directly using KMS customer managed keys for S3 encryption is more straightforward and meets the requirements with less effort. Also, data encrypted with customer keys needs to be decrypted first before uploading, adding further complexity and possible security risk. The first step in this answer is also incorrect - the data should be encrypted within S3.

Overall Explanation:

The best solution is to create a KMS customer managed key, enable automatic key rotation, and then set the S3 bucket's default encryption to use this KMS key. This provides

encryption at rest and automatic yearly key rotation with the least operational overhead. SSE-S3 does not guarantee yearly rotation and manual rotation increases overhead. Importing customer key material is unnecessarily complex.

[View Discussions](#)[Ask AI](#)

Question 48

Not Attempted

Security



A company is migrating applications from an on-premises Microsoft Active Directory, managed by the company, to AWS. The applications are deployed across multiple AWS accounts managed centrally using AWS Organizations. The security team requires a single sign-on (SSO) solution for all AWS accounts while continuing to manage users and groups in the on-premises Active Directory. Which solution meets these requirements?

- ☐ Create an Enterprise Edition Active Directory in AWS Directory Service for Microsoft Active Directory. Configure the Active Directory to be the identity source for AWS IAM Identity Center.

Creating a new Active Directory requires migrating users and groups, failing to meet the requirement of continuing to use the on-premises AD.

- ☒ Enable AWS IAM Identity Center. Configure a two-way forest trust relationship to connect the company's self-managed Active Directory with IAM Identity Center by using AWS Directory Service for Microsoft Active Directory.

This solution allows IAM Identity Center to integrate with the existing on-premises AD for authentication and authorization, providing SSO across AWS accounts while retaining the on-premises AD as the primary identity source.

- ☐ Use AWS Directory Service and create a two-way trust relationship with the company's self-managed Active Directory.

While this establishes trust, it does not provide the SSO functionality needed for the AWS accounts. IAM Identity Center is needed for SSO.

- ☐ Deploy an identity provider (IdP) on Amazon EC2. Link the IdP as an identity source within AWS IAM Identity Center.

While feasible, deploying an IdP on EC2 introduces unnecessary overhead and complexity compared to using AWS Directory Service for Microsoft Active Directory with IAM Identity Center.

Overall Explanation:

The correct answer is to enable AWS IAM Identity Center and configure a two-way forest trust relationship using AWS Directory Service for Microsoft Active Directory. This allows seamless integration with the existing on-premises Active Directory, providing SSO across multiple AWS accounts without migrating users or groups.

[View Discussions](#)[Ask AI](#)

Question 49

Not Attempted

Database



A company is planning to deploy its application on an Amazon Aurora PostgreSQL Serverless v2 cluster. The application will receive large amounts of traffic. The company wants to optimize the storage performance of the cluster as the load on the application increases and minimize I/O costs.

- ☐ ☒ Configure the cluster to use the Aurora Standard storage configuration.
Aurora Standard storage may be cheaper for low I/O workloads, but for high traffic applications, the I/O costs can quickly become prohibitive. It charges for I/O operations.
- ☐ ☒ Configure the cluster storage type as Provisioned IOPS.
Provisioned IOPS is not an applicable storage configuration for Aurora.
- ☐ ☒ Configure the cluster storage type as General Purpose.
General Purpose is not an applicable storage configuration for Aurora.
- ☒ ☐ Configure the cluster to use the Aurora I/O-Optimized storage configuration.
Aurora I/O-Optimized is designed for I/O-intensive applications. While the per GB-month cost may be higher, it eliminates charges for read and write I/O operations, potentially leading to significant cost savings for high-traffic applications.

Overall Explanation:

Aurora I/O-Optimized provides improved price performance for I/O-intensive applications by eliminating charges for I/O operations. Given the large amounts of traffic, this option is the most cost-effective because the business wants to optimize storage performance of the cluster as the load on the application increases. Aurora only has Standard and I/O-

Optimized storage options. Provisioned IOPS and General Purpose storage types are not valid storage configurations for Aurora.

[View Discussions](#)[Ask AI](#)

Question 50

Not Attempted

Security



A financial services company operating on AWS needs to demonstrate compliance with industry standards like NIST and PCI DSS to its third-party auditors. The company has hundreds of AWS accounts within a single AWS Organizations setup. They require a solution to consistently monitor the state of their security controls across all accounts and provide evidence of implementation and proper functioning to the auditors. Which AWS service will best satisfy these requirements?

- ☐ Designate one account as the Amazon Inspector delegated administrator account from the Organizations management account. Integrate Inspector with Organizations to discover and scan resources across all AWS accounts. Enable Inspector industry standards for NIST and PCI DSS.

Inspector primarily focuses on vulnerability assessments, not continuous compliance monitoring against industry standards across an entire organization.

- ☐ Designate one account as the Amazon GuardDuty delegated administrator account from the Organizations management account. In the designated GuardDuty administrator account, enable GuardDuty to protect all member accounts. Enable GuardDuty industry standards for NIST and PCI DSS.

GuardDuty is a threat detection service and does not provide compliance monitoring or reporting capabilities for NIST and PCI DSS.

- ☐ Configure an AWS CloudTrail organization trail in the Organizations management account. Designate one account as the compliance account. Enable CloudTrail security standards for NIST and PCI DSS in the compliance account.

CloudTrail primarily logs API calls; it doesn't provide a compliance dashboard or assess configurations against NIST and PCI DSS standards.

- ☒ Designate one account as the AWS Security Hub delegated administrator account from the Organizations management account. In the designated Security Hub administrator account, enable Security Hub for all member accounts. Enable Security Hub standards for NIST and PCI DSS.

Security Hub integrates with AWS Organizations, supports multiple compliance standards like NIST and PCI DSS, and allows for centralized monitoring of security posture across all accounts.

Overall Explanation:

AWS Security Hub is the most appropriate service. By designating a delegated administrator account, Security Hub can be enabled for all member accounts within AWS Organizations. It supports enabling security standards like NIST and PCI DSS, providing a centralized view of compliance status and enabling the company to demonstrate the implementation and functioning of security controls to auditors.

[View Discussions](#)[Ask AI](#)

Question 51

Not Attempted

Storage



A company uses an Amazon S3 bucket as its data lake storage platform. The S3 bucket contains a massive amount of data that is accessed randomly by multiple teams and hundreds of applications. The company wants to reduce the S3 storage costs and provide immediate availability for frequently accessed objects. What is the MOST operationally efficient solution that meets these requirements?

- ☒ Create an S3 Lifecycle rule to transition objects to the S3 Intelligent-Tiering storage class.
S3 Intelligent-Tiering automatically stores objects in access tiers optimized for frequent, infrequent, and rarely accessed data, reducing costs and providing immediate availability.
- ☐ Store objects in Amazon S3 Glacier. Use S3 Select to provide applications with access to the data.
S3 Glacier is for archival data and S3 Select, while useful, does not provide immediate availability as required.
- ☐ Use data from S3 storage class analysis to create S3 Lifecycle rules to automatically transition objects to the S3 Standard-Infrequent Access (S3 Standard-IA) storage class.
While this approach is valid, it requires manual configuration and analysis, which is less operationally efficient than S3 Intelligent-Tiering.
- ☐ Transition objects to the S3 Standard-Infrequent Access (S3 Standard-IA) storage class. Create an AWS Lambda function to transition objects to the S3 Standard storage class.

when they are accessed by an application.

This approach introduces operational overhead with managing a Lambda function and is less efficient than the automated tiering provided by S3 Intelligent-Tiering.

Overall Explanation:

S3 Intelligent-Tiering is the most operationally efficient solution because it automatically manages storage tiers based on access patterns, ensuring immediate availability for frequently accessed objects while reducing storage costs for infrequently accessed objects without manual intervention.

[View Discussions](#)[Ask AI](#)

Question 52

Not Attempted

Networking



A company's dynamic website is hosted using on-premises servers in the United States. The company is launching its product in Europe, and it wants to optimize site loading times for new European users. The site's backend must remain in the United States. The product is being launched in a few days, and an immediate solution is needed. What should the solutions architect recommend?

- ☐ Launch an Amazon EC2 instance in us-east-1 and migrate the site to it.
Launching a single EC2 instance, even in a US region, will not effectively reduce latency for European users and migrating the site takes time.
- ☐ Move the website to Amazon S3. Use Cross-Region Replication between Regions.
Amazon S3 is designed for static content and doesn't provide the server-side processing capabilities needed for a dynamic website. Also, this migration would take too long.
- ☒ Use Amazon CloudFront with a custom origin pointing to the on-premises servers.
Amazon CloudFront, a CDN, caches content at edge locations closer to users, reducing latency. Using a custom origin allows it to point to the existing on-premises servers without requiring a full migration. This is also the quickest solution.
- ☐ Use an Amazon Route 53 geoproximity routing policy pointing to on-premises servers.
While Route 53 geoproximity routing directs users to the closest server, it doesn't cache content like a CDN, so European users would still experience latency accessing the US-based on-premises servers.

Overall Explanation:

Amazon CloudFront is the best solution because it's a CDN that can cache static and dynamic content at edge locations closer to European users, thereby reducing latency. The custom origin feature allows it to seamlessly integrate with the existing on-premises infrastructure without requiring a complete migration, making it the fastest option to implement. The other options involve full migration and would take longer to implement.

[View Discussions](#)[Ask AI](#)

Question 53

Not Attempted

Database



A social media company has a dataset of 5 TB containing 1 million user profiles and 10 million connections represented as many-to-many relationships. They need to efficiently identify mutual connections between users up to five degrees of separation. Which AWS service is best suited for this task, ensuring high performance and scalability?

- ⊗ Amazon S3 with Amazon Athena using SQL JOIN queries.

While S3 is suitable for storing large datasets and Athena allows querying data in S3, using SQL JOIN queries for finding connections up to five levels will be inefficient and slow due to the complex relationships.

- ✓ Amazon Neptune, storing the data as a graph database with nodes and edges.

Amazon Neptune is a graph database service designed for handling highly connected data. It can efficiently traverse relationships and find connections up to multiple levels, making it ideal for this use case.

- ⊗ Amazon S3 with Amazon QuickSight for visualizing connections.

QuickSight is a business intelligence service for visualization, not for efficiently querying highly connected data for multiple degrees of separation.

- ⊗ Amazon RDS using multiple tables with SQL JOIN queries.

Similar to Athena, using a relational database like RDS and SQL JOIN queries for finding connections up to five levels will be inefficient and slow due to the complex relationships.

Overall Explanation:

Amazon Neptune is the best choice because it's a graph database specifically designed for efficiently managing and querying highly connected data like social networks. It allows for fast traversal of relationships to find mutual connections up to five levels. Relational databases and S3-based solutions with SQL JOIN queries or visualization tools will be less efficient for this type of complex relationship analysis. Neptune automatically scales storage and compute resources based on workload demands, ensuring optimal performance even as the dataset grows over time.

<https://docs.aws.amazon.com/neptune/latest/userguide/notebooks-visualization.html>

 View Discussions

 Ask AI

Question 54

Not Attempted

Networking



A company needs to establish a secure and cost-effective connection between their on-premises data center and their AWS environment. The connection requires low bandwidth and will only handle a small amount of traffic. The setup also needs to be implemented as quickly as possible. Which of the following options is the MOST suitable for this scenario?

☐ Implement a client VPN.

A client VPN is more suited for individual users connecting to the network, not for a site-to-site connection.

☐ Implement AWS Direct Connect.

AWS Direct Connect is designed for high-bandwidth, low-latency connections and is a more expensive option, making it unsuitable for low-bandwidth requirements.

☐ Implement a bastion host on Amazon EC2.

A bastion host provides secure access to EC2 instances within a VPC but doesn't establish a site-to-site connection between the on-premises network and AWS.

☒ Implement an AWS Site-to-Site VPN connection.

An AWS Site-to-Site VPN provides a secure, cost-effective, and quick way to connect an on-premises network to AWS. It is ideal for low-bandwidth and low-traffic scenarios.

Overall Explanation:

AWS Site-to-Site VPN is the most cost-effective and appropriate solution for establishing a secure connection between an on-premises environment and AWS when low bandwidth and quick setup are required. Direct Connect is overkill for low bandwidth. A client VPN is for individual users, and a bastion host does not provide the desired site-to-site connection. Therefore, Site-to-Site VPN is the best choice.

[View Discussions](#)[Ask AI](#)

Question 55

Not Attempted

Security



A company is migrating its on-premises SFTP file transfer solution to AWS to improve scalability and reduce costs using Amazon S3. Employees need to use their existing on-premises Microsoft Active Directory (AD) credentials to access the new solution. The company wants to maintain its current authentication and file access mechanisms with minimal operational overhead. Which solution is the MOST suitable?

- ☐ Configure an S3 File Gateway. Create SMB file shares on the file gateway that use the existing Active Directory to authenticate.

S3 File Gateway is primarily for presenting S3 data as a file share to on-premises applications, not for direct SFTP access with AD authentication. It also involves more infrastructure to manage.

- ☐ Configure an Auto Scaling group with Amazon EC2 instances to run an SFTP solution. Configure the group to scale up at 60% CPU utilization.

This solution requires managing EC2 instances, patching, and the SFTP software itself, leading to high operational overhead. It also doesn't natively integrate with AD for authentication.

- ☒ Create an AWS Transfer Family server with SFTP endpoints. Choose the AWS Directory Service option as the identity provider. Use AD Connector to connect the on-premises Active Directory.

AWS Transfer Family simplifies SFTP server management and natively integrates with AWS Directory Service. Using AD Connector allows leveraging existing on-premises AD credentials for authentication with minimal operational overhead.

- ⊗ Create an AWS Transfer Family SFTP endpoint. Configure the endpoint to use the AWS Directory Service option as the identity provider to connect to the existing Active Directory.

This option misses a critical step. Transfer Family requires a *server* to be created before an endpoint can be configured. Without creating the server, no SFTP service is enabled.

Overall Explanation:

AWS Transfer Family provides a managed SFTP service that integrates with Amazon S3 and supports authentication via AWS Directory Service. By using AD Connector, the solution allows employees to use their existing on-premises Active Directory credentials for SFTP access. This option eliminates the need for managing servers and minimizes operational overhead compared to alternatives. Option C correctly explains the need to create Transfer Family *server* and use AD Connector.

 View Discussions

 Ask AI

Question 56

Not Attempted

Serverless



A company is designing an event-driven order processing system. Upon order creation, multiple independent validation steps are required, each performed by an idempotent AWS Lambda function. Each validation step only requires a specific subset of the order event data. The company needs to ensure each Lambda function has access only to its required data subset. Furthermore, the system components should be loosely coupled to accommodate future business changes. Which solution best meets these requirements?

- ⊗ Create an Amazon Simple Queue Service (Amazon SQS) queue for each validation step. Create a new Lambda function to transform the order data to the format that each validation step requires and publish the messages to the appropriate SQS queues. Subscribe each validation step Lambda function to its corresponding SQS queue. This solution requires an additional Lambda function for data transformation and introduces complexity with multiple SQS queues. It's a valid approach but not the most efficient.
- ⊗ Create an Amazon Simple Notification Service (Amazon SNS) topic. Subscribe the validation step Lambda functions to the SNS topic. Use message body filtering to send only the required data to each subscribed Lambda function.

SNS message filtering cannot manipulate messages; it can only discard or forward the FULL message based on attribute matching. It doesn't allow for sending only a subset of the data.

- ✔ Create an Amazon EventBridge event bus. Create an event rule for each validation step. Configure the input transformer to send only the required data to each target validation step Lambda function.

Amazon EventBridge with input transformation allows for routing events based on rules and transforming the event payload to send only the required data subset to each Lambda function. This promotes loose coupling and efficient data handling.

- ✘ Create an Amazon Simple Queue Service (Amazon SQS) queue. Create a new Lambda function to subscribe to the SQS queue and to transform the order data to the format that each validation step requires. Use the new Lambda function to perform synchronous invocations of the validation step Lambda functions in parallel on separate threads.

This approach introduces tighter coupling through synchronous invocations and requires an additional Lambda function for transformation and orchestration, making it less ideal than EventBridge.

Overall Explanation:

Amazon EventBridge provides the best solution because it allows for event-driven architecture and input transformation. The input transformer feature enables sending only the necessary data subset to each Lambda function, satisfying the requirement of providing each validation step with only the information it needs. Furthermore, EventBridge facilitates loose coupling between components, accommodating future business changes more effectively. Option A requires an extra Lambda, option B cannot transform the message to only include certain attributes, and option D uses synchronous invocations which creates tight coupling. Therefore, EventBridge is the most suitable and efficient approach.

 View Discussions

 Ask AI

Question 57

Not Attempted

Database



A company is migrating a three-tier application to AWS. The application utilizes a MySQL database. Historically, users have reported performance degradation when creating new entries due to the generation of real-time reports during business hours. Which solution will optimize

the application's performance in AWS, specifically addressing the reporting-related performance bottlenecks?

- ⊗ Import the data into an Amazon DynamoDB table with provisioned capacity. Refactor the application to use DynamoDB for reports.

DynamoDB is a NoSQL database suitable for high-throughput reads and writes. While it could handle report generation, refactoring the entire application to use DynamoDB solely for reports is a significant undertaking and might not be the most efficient solution for the described problem.

- ⊗ Create the database on a compute optimized Amazon EC2 instance. Ensure compute resources exceed the on-premises database.

While increasing compute resources might improve overall database performance, it doesn't isolate the reporting workload, meaning that the performance issues related to real-time reports impacting new entry creation are unlikely to be fully resolved.

- ✔ Create an Amazon Aurora MySQL Multi-AZ DB cluster with multiple read replicas. Configure the application to use the reader endpoint for reports.

This is the most suitable solution. Aurora MySQL read replicas allow read-only queries to be offloaded from the primary database instance, preventing report generation from impacting the performance of write operations (new entry creation). The reader endpoint distributes read traffic across multiple replicas, further enhancing performance.

- ⊗ Create an Amazon Aurora MySQL Multi-AZ DB cluster. Configure the application to use the backup instance of the cluster as an endpoint for the reports.

While Aurora Multi-AZ clusters have a standby instance for failover, it's not intended to be used as a read replica in the same way dedicated read replicas are. Furthermore, using the backup instance directly for reports is not a supported or recommended practice, as it can interfere with the failover process and potentially lead to data inconsistency.

Overall Explanation:

The most effective solution is to create an Amazon Aurora MySQL Multi-AZ DB cluster with multiple read replicas. By configuring the application to use the reader endpoint for report generation, the read-heavy report queries are offloaded from the primary database instance. This prevents the performance impact on write operations, such as creating new entries, addressing the reported performance issues. Options A and B are not the best choices as they involve unnecessary overhead. Using a backup instance is also not recommended.

Question 58

Not Attempted

Networking



A company is expanding a secure on-premises network to the AWS Cloud by using an AWS Direct Connect connection. The on-premises network has no direct internet access. An application that runs on the on-premises network needs to access an Amazon S3 bucket in the MOST cost-effective manner. Which solution meets these requirements?

- ☐ Create a public virtual interface (VIF). Route the AWS traffic over the public VIF.

This option is incorrect because a public VIF requires internet access, which the on-premises network does not have.

- ☐ Create a VPC and a NAT gateway. Route the AWS traffic from the on-premises network to the NAT gateway.

This option is incorrect because NAT gateway requires internet access for traffic routing, which is not available for the on-premises network.

- ☒ Create a VPC and an Amazon S3 interface endpoint. Route the AWS traffic from the on-premises network to the S3 interface endpoint.

This is the correct answer. Using an S3 interface endpoint allows the on-premises network to access S3 privately, without traversing the internet, and is a cost-effective solution.

- ☐ Create a VPC peering connection between the on-premises network and Direct Connect. Route the AWS traffic over the peering connection.

VPC peering is used for connectivity between VPCs and does not provide access to AWS public services such as Amazon S3.

Overall Explanation:

The MOST cost-effective solution is to create a VPC and an Amazon S3 interface endpoint. This allows private access to S3 without the need for internet access, which aligns with the constraint that the on-premises network has no direct internet access. Public VIF requires internet access and NAT Gateway also requires internet access. VPC peering doesn't provide access to public AWS Services.

[View Discussions](#)[Ask AI](#)

Question 59

Not Attempted

Networking



A company hosts its website using an Auto Scaling group of Amazon EC2 instances in a single AWS Region. The website does not rely on a database. The company is expanding and has deployed the website to a second Region. They want to distribute traffic across both Regions to handle increased traffic and ensure disaster recovery. The solution should prevent serving traffic from a Region where the website is unhealthy. Which policy or resource should the company use to meet these requirements?

- ☐ An Amazon Route 53 simple routing policy
A simple routing policy does not support health checks, so it cannot ensure traffic is only served from healthy Regions.
- ☒ An Amazon Route 53 multivalue answer routing policy
A multivalue answer routing policy allows configuring Amazon Route 53 to return multiple values (e.g., IP addresses) in response to DNS queries. It also supports health checks, returning only values for healthy resources. This improves availability and load balancing.
- ☐ An Application Load Balancer in one Region with a target group that specifies the EC2 instance IDs from both Regions
Application Load Balancers (ALBs) operate within a single Region. Specifying EC2 instances from different Regions in a single ALB's target group is not a valid configuration.
- ☐ An Application Load Balancer in one Region with a target group that specifies the IP addresses of the EC2 instances from both Regions
While ALBs can use IP addresses as targets, they still operate within a single Region. Distributing traffic across Regions using a single ALB is not possible. Also, this approach wouldn't provide native health checking across regions.

Overall Explanation:

Amazon Route 53 multivalue answer routing policy is the best solution. It allows returning multiple IP addresses (from different Regions) in response to DNS queries, and it supports health checks to ensure that only healthy instances/Regions receive traffic. Simple routing policies lack health checks, and Application Load Balancers operate within a single Region, making them unsuitable for cross-region traffic distribution and failover.

View Discussions

Ask AI

Question 60

Not Attempted

Storage



A company's applications run on Amazon EC2 instances backed by Amazon EBS. These instances use the latest Amazon Linux release. Employees are experiencing availability issues when storing and retrieving files that are 25 GB or larger. The company requires a solution that avoids file transfers between EC2 instances and ensures file availability across many EC2 instances in multiple Availability Zones. Which solution best meets these requirements?

- ☐ Migrate all files to an Amazon S3 bucket and instruct employees to access the files from the S3 bucket.

While S3 provides durable object storage, it might not be the most efficient for frequent file access directly from the EC2 instances. It also doesn't inherently address the 'available across many EC2 instances' requirement without additional configuration.

- ☐ Take a snapshot of the existing EBS volume. Mount the snapshot as an EBS volume across the EC2 instances. Instruct the employees to access the files from the EC2 instances.

EBS volumes are region-locked. Mounting a snapshot as a volume on multiple EC2 instances across Availability Zones would require creating multiple volumes from the snapshot, increasing management overhead and not providing a shared file system.

- ☒ Mount an Amazon Elastic File System (Amazon EFS) file system across all the EC2 instances. Instruct the employees to access the files from the EC2 instances.

Amazon EFS is designed to be mounted concurrently by multiple EC2 instances across multiple Availability Zones. This fulfills the requirement of shared access and availability across different AZs without manual data transfer.

- ☐ Create an Amazon Machine Image (AMI) from the EC2 instances. Configure new EC2 instances from the AMI that use an instance store volume. Instruct the employees to access the files from the EC2 instances.

Instance store volumes are ephemeral and the data is lost when the instance is stopped or terminated. Also, sharing data across Availability Zones wouldn't be inherently addressed.

Overall Explanation:

Amazon EFS provides a shared file system that can be mounted concurrently by multiple EC2 instances, addressing the requirement for file availability across many EC2 instances and multiple Availability Zones. It eliminates the need for file transfers between instances and maintains availability when dealing with large files.

[View Discussions](#)[Ask AI](#)

Question 61

Not Attempted

Security



A company is running a highly sensitive application on Amazon EC2 backed by an Amazon RDS database. Compliance regulations mandate that all personally identifiable information (PII) be encrypted at rest. Which solution should a solutions architect recommend to meet this requirement with the LEAST amount of changes to the infrastructure?

- ☐ Deploy AWS Certificate Manager to generate certificates. Use the certificates to encrypt the database volume.
AWS Certificate Manager is used for managing SSL/TLS certificates for securing network communications, not for encrypting data at rest on database volumes.
- ☐ Deploy AWS CloudHSM, generate encryption keys, and use the keys to encrypt database volumes.
CloudHSM provides hardware security modules in the AWS cloud. While this can be used for encryption, it involves more infrastructure changes and complexity than necessary for this requirement.
- ☐ Configure SSL encryption using AWS Key Management Service (AWS KMS) keys to encrypt database volumes.
SSL encryption secures data in transit, not data at rest. While using KMS is appropriate, SSL is not the correct mechanism for encrypting the database volumes themselves.
- ☒ Configure Amazon Elastic Block Store (Amazon EBS) encryption and Amazon RDS encryption with AWS Key Management Service (AWS KMS) keys to encrypt instance and database volumes.
This option provides encryption at rest for both the EC2 instance's EBS volumes and the RDS database volumes, using KMS for key management. It is the least disruptive approach as it leverages existing AWS features.

Overall Explanation:

Option D provides the most straightforward way to meet the encryption-at-rest compliance requirement with minimal changes to the existing infrastructure. RDS relies on EBS volumes for storage, therefore encrypting both EBS and RDS ensures comprehensive protection. Using KMS allows centralized key management.

[View Discussions](#)[Ask AI](#)

Question 62

Not Attempted

Networking



A company runs an AWS Lambda function in private subnets within a VPC. These subnets route outbound traffic through an EC2 NAT instance to access the internet. The Lambda function processes data and uploads the resulting objects to Amazon S3. Intermittently, the Lambda function experiences timeouts during the object upload process, attributed to network saturation on the NAT instance. The company aims to enable the Lambda function to access Amazon S3 without routing traffic through the internet. Which solution satisfies these requirements?

- ☐ Replace the EC2 NAT instance with an AWS managed NAT gateway.
Replacing the EC2 NAT instance with a NAT gateway will not prevent the Lambda function from traversing the internet to access S3.
- ☐ Increase the size of the EC2 NAT instance in the VPC to a network optimized instance type.
Increasing the size of the EC2 NAT instance might alleviate the saturation temporarily, but it does not prevent the Lambda function from traversing the internet to access S3. The problem statement explicitly states that the company wants to avoid internet traversal.
- ☒ Provision a gateway endpoint for Amazon S3 in the VPC. Update the route tables of the subnets accordingly.
A gateway endpoint allows direct access to S3 from within the VPC, without needing to route traffic through the internet. By updating the subnet route tables to use the gateway endpoint, the Lambda function can access S3 directly.
- ☐ Provision a transit gateway. Place transit gateway attachments in the private subnets where the Lambda function is running.
A transit gateway is primarily used to connect multiple VPCs and on-premises networks. While it could be used, it is not the most straightforward or cost-effective solution for accessing S3 from a Lambda function in private subnets without traversing the internet. Gateway endpoints are the preferred way to achieve this.

Overall Explanation:

Provisioning a gateway endpoint for Amazon S3 allows the Lambda function in the private subnets to directly access S3 without routing traffic through the internet or a NAT instance.

Updating the route tables ensures that traffic destined for S3 uses the endpoint. This eliminates the saturation issue on the NAT instance and fulfills the requirement of avoiding internet traversal. NAT Gateways will still require internet access and Transit Gateways are used to connect VPCs, so they will not solve the problem as directly and cost-effectively as a Gateway Endpoint.

[View Discussions](#)[Ask AI](#)

Question 63

Not Attempted

[Compute](#)

A company aims to minimize costs for its existing three-tier web architecture hosted on Amazon EC2 instances across development, test, and production environments. The instances show average CPU utilization of 30% during peak hours and 10% during off-peak hours. Production instances must run 24/7. Development and test instances are required for at least 8 hours daily, with plans to automate their shutdown when not in use. Which EC2 instance purchasing option provides the MOST cost-effective solution for the company's needs?

- ☐ Use Spot Instances for the production EC2 instances. Use Reserved Instances for the development and test EC2 instances.
Spot Instances are not suitable for production environments requiring continuous availability. Reserved Instances may not be optimal for Dev/Test as these are not used 24/7.
- ☒ Use Reserved Instances for the production EC2 instances. Use On-Demand Instances for the development and test EC2 instances.
Reserved Instances are cost-effective for production instances that run continuously. On-Demand Instances offer flexibility for development and test instances with variable usage patterns.
- ☐ Use Spot Blocks for the production EC2 instances. Use Reserved Instances for the development and test EC2 instances.
Spot Blocks are no longer available as an option. Reserved Instances may not be optimal for Dev/Test as these are not used 24/7.
- ☐ Use On-Demand Instances for the production EC2 instances. Use Spot Blocks for the development and test EC2 instances.
On-Demand Instances for production will be the most expensive option. Spot Blocks are no longer available.

Overall Explanation:

The most cost-effective solution is to utilize Reserved Instances for the production environment, which operates continuously. This provides significant savings compared to On-Demand pricing. For the development and test environments, On-Demand Instances are ideal due to their variable usage patterns and the ability to stop them when not needed. This avoids paying for idle resources, ensuring cost optimization.

[View Discussions](#)[Ask AI](#)

Question 64

Not Attempted

Networking



A global news company uses AWS to host its broadcast system. Reporters worldwide send live broadcasts using RTMP (Real-Time Messaging Protocol) from their phones to the system. A solutions architect needs to design a solution that allows reporters to send the highest quality streams with accelerated TCP connections back to the broadcast system. Which AWS service should the solutions architect use?

☐ Amazon CloudFront

CloudFront is a CDN primarily used for caching and delivering static and dynamic web content. While it can accelerate HTTP(S) traffic, it's not the best solution for general TCP acceleration required for RTMP streaming.

☒ AWS Global Accelerator

AWS Global Accelerator provides static IP addresses that serve as entry points for applications. It optimizes the network path to improve TCP and UDP performance, making it suitable for accelerating RTMP streams from reporters worldwide.

☐ AWS Client VPN

AWS Client VPN provides secure remote access to AWS resources. While it offers secure connections, it doesn't inherently accelerate TCP connections for live streaming.

☐ Amazon EC2 instances and AWS Elastic IP addresses

Using EC2 instances with Elastic IP addresses provides static IPs, but it doesn't offer the global network optimization and TCP acceleration provided by Global Accelerator.

Overall Explanation:

AWS Global Accelerator is the most suitable choice because it is designed to optimize TCP and UDP traffic globally, providing accelerated and reliable connections back to the broadcast system. It uses the AWS global network to route traffic efficiently, improving the quality and reliability of live streams from reporters around the world.

[View Discussions](#)[Ask AI](#)

Question 65

Not Attempted

Storage



A company uses Amazon EC2 instances and Amazon Elastic Block Store (Amazon EBS) to run its self-managed database. The company has 350 TB of data spread across all EBS volumes. The company takes daily EBS snapshots and keeps the snapshots for 1 month. The daily change rate is 5% of the EBS volumes. Because of new regulations, the company needs to keep the monthly snapshots for 7 years. The company needs to change its backup strategy to comply with the new regulations and to ensure that data is available with minimal administrative effort. Which solution will meet these requirements MOST cost-effectively?

- ☐ Keep the daily snapshot in the EBS snapshot standard tier for 1 month. Copy the monthly snapshot to Amazon S3 Glacier Deep Archive with a 7-year retention period.
This option involves copying to S3 Glacier Deep Archive. While Glacier is cost-effective for long-term storage, the initial transfer and potential restores can add complexity and administrative overhead.
- ☒ Continue with the current EBS snapshot policy. Add a new policy to move the monthly snapshot to Amazon EBS Snapshots Archive with a 7-year retention period.
EBS Snapshots Archive is designed for long-term EBS snapshot retention and provides a balance between cost and accessibility. Using it in conjunction with the existing snapshot policy offers a straightforward solution.
- ☐ Keep the daily snapshot in the EBS snapshot standard tier for 1 month. Keep the monthly snapshot in the standard tier for 7 years. Use incremental snapshots.
Keeping snapshots in the standard tier for 7 years would be prohibitively expensive, making this option not cost-effective.

- ⊗ Keep the daily snapshot in the EBS snapshot standard tier. Use EBS direct APIs to take snapshots of all the EBS volumes every month. Store the snapshots in an Amazon S3 bucket in the Infrequent Access tier for 7 years.

Using EBS direct APIs and storing in S3 Infrequent Access involves more management overhead than using EBS Snapshot Archive and might not be as cost-effective.

Overall Explanation:

The most cost-effective solution that meets the requirements is to continue the current EBS snapshot policy and add a new policy to move the monthly snapshot to Amazon EBS Snapshots Archive with a 7-year retention period. This solution leverages a feature designed for long-term snapshot storage and minimizes administrative effort. While S3 Glacier Deep Archive is cheaper, the effort to copy data there and potentially restore it adds complexity.

 [View Discussions](#)

 [Ask AI](#)

[← Back to Exam](#)



Marketing

[Affiliate Program](#)

[Unlimited Access](#)

[Business Unlimited Access](#)

Support

[Contact Us](#)

Company

[Udemy Alternative](#)

[Whizlabs Alternative](#)

[TutorialsDojo Alternative](#)

Legal

[Terms of service](#)

team@itcertifications.io

[Privacy policy](#)

[Refund policy](#)

[Shipping policy](#)



© 2026, ITcertifications Technologies. All rights reserved.